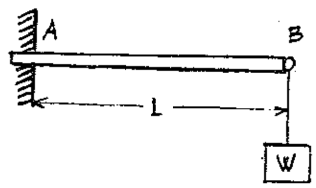
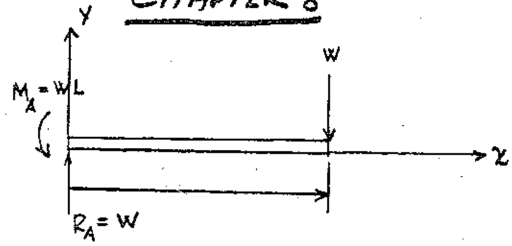
(8-1) (a)

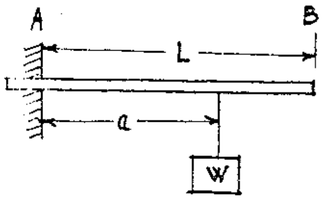


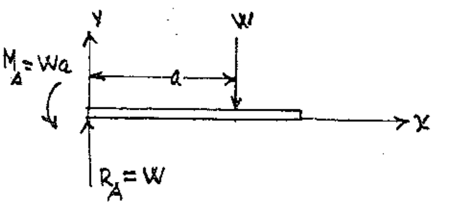


$$EI\frac{d^2v}{dx^2} = M_b = R_A x - M_A = W(x-L) \qquad \text{B.C. } x=0 \quad v=0\;;\; \frac{dv}{dx}=0$$

$$v = \frac{W}{2EI}\left(\frac{x^3}{3} - Lx^2\right)$$

(8-1) (b)

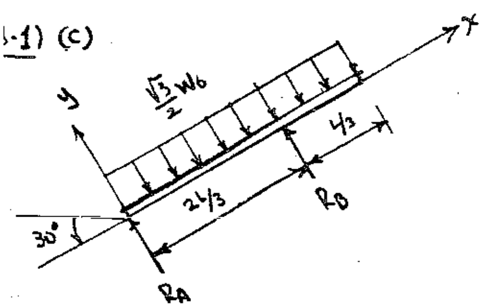




$$EI\frac{d^2v}{dx^2} = M_b = R_A x - W\langle x-a\rangle^1 - Wa \qquad \text{B.C. } x=0, \quad v=0\;;\; \frac{dv}{dx}=0$$

$$v = \frac{W}{6EI}\left(x^3 - \langle x-a\rangle^3 + 6ax^2\right)$$

(8-1) (c)



From Prob. (3-23c)

$$M_B(x) = R_A\langle x\rangle^1 - \frac{\sqrt{3}}{2}W_0\langle x\rangle^2 + R_B\langle x - \tfrac{2L}{3}\rangle^1 \qquad R_A = \frac{\sqrt{3}W_0 L}{8}$$

$$M_B(x) = EI\frac{d^2v}{dx^2} \qquad R_B = \frac{3\sqrt{3}}{8}W_0 L$$

$$\therefore\; EI\frac{dv}{dx} = \frac{R_A}{2}\langle x\rangle^2 - \frac{\sqrt{3}}{6}W_0\langle x\rangle^3 + \frac{R_B}{2}\langle x - \tfrac{2L}{3}\rangle^2 + C_1$$

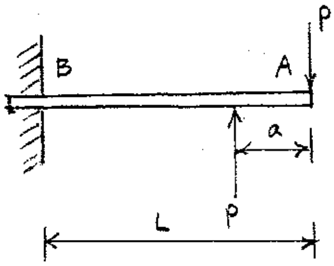
$$EI\,v = \frac{R_A}{6}\langle x\rangle^3 - \frac{\sqrt{3}}{24}W_0\langle x\rangle^4 + \frac{R_B}{6}\langle x - \tfrac{2L}{3}\rangle^3 + C_1 x + C_2$$

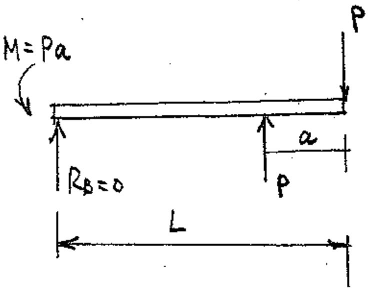
$$x=0\;;\; v=0 \qquad C_2 = 0$$

$$x = \frac{2L}{3} \quad v=0 \qquad C_1 = \frac{\sqrt{3}}{24}W_0\left(\frac{2L}{3}\right)^3 - \frac{R_A}{6}\left(\frac{2L}{3}\right)^2$$

$$\therefore\; v = \frac{\sqrt{3}W_0}{48EI}\left\{L\langle x\rangle^3 - 2\langle x\rangle^4 + 3L\langle x - \tfrac{2L}{3}\rangle^3 + 2\left(\tfrac{2L}{3}\right)^3 x - L\left(\tfrac{2L}{3}\right)^2 x\right\}$$

8-1 (d)





$F_y = 0$

$R_B + P - P = 0$

$R_B = 0$

$\Sigma M = 0$

$-M_B + PL - P(L-a) = 0$

$M_B = Pa$

$$M_B = -Pa + P\langle x - L + a\rangle^1 = EI\frac{d^2v}{dx^2}$$

$$EI\frac{dv}{dx} = -Pax + \frac{P}{2}\langle x - L + a\rangle^2 + C_1$$

$$\left(\frac{dv}{dx}\right)_{x=0} = 0 \qquad C_1 = 0$$

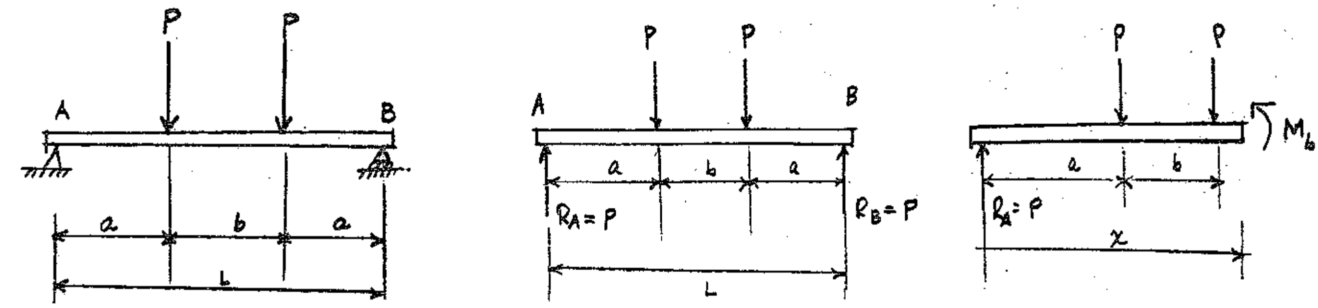
$$\therefore \frac{dv}{dx} = \frac{P}{EI}\left[-ax + \frac{\langle x - L + a\rangle^2}{2}\right]$$

$$v = \frac{P}{EI}\left[-\frac{ax^2}{2} + \frac{\langle x - L + a\rangle^3}{6}\right] + C_2$$

$$(v)_{x=0} = 0 \qquad C_2 = 0$$

$$v = \frac{P}{EI}\left[-\frac{ax^2}{2} + \frac{\langle x - L + a\rangle^3}{6}\right]$$

8.1(e)



$$R_A = P = R_B$$

$$M_b = Px - P\langle x-a\rangle^1 - P\langle x-a-b\rangle^1 = EI\frac{d^2v}{dx^2}$$

$$EI\frac{dv}{dx} = \frac{Px^2}{2} - \frac{P}{2}\langle x-a\rangle^2 - \frac{P}{2}\langle x-a-b\rangle^2 + C_1$$

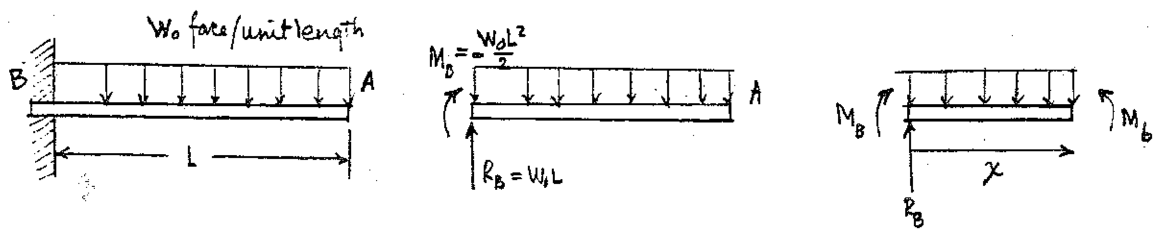
$$EI\,v = \frac{Px^3}{6} - \frac{P}{6}\langle x-a\rangle^3 - \frac{P}{6}\langle x-a-b\rangle^3 + C_1x + C_2$$

$$(v)_{x=0} = 0 \qquad C_2 = 0$$

$$(v)_{x=L} = 0 \qquad C_1 = \frac{P}{6EIL}\left[-L^3 + (a+b)^3 + a^3\right]$$

$$\boxed{v = \frac{P}{6EI}\left[(x^3 - L^2x) - \langle x-a\rangle^3 - \langle x-a-b\rangle^3 + (a+b)^3\frac{x}{L} + a^3\frac{x}{L}\right]}$$

8.1(f)



$$M_b = M_B + R_Bx = -\frac{W_0L^2}{2} + W_0Lx - \frac{Wx^2}{2} = EI\frac{d^2v}{dx^2}$$

$$EI\frac{dv}{dx} = -\frac{W_0L^2x}{2} + \frac{W_0Lx^2}{2} - \frac{Wx^3}{6} + C_1$$

$$\left(\frac{dv}{dx}\right)_{x=0} = 0 \qquad C_1 = 0$$

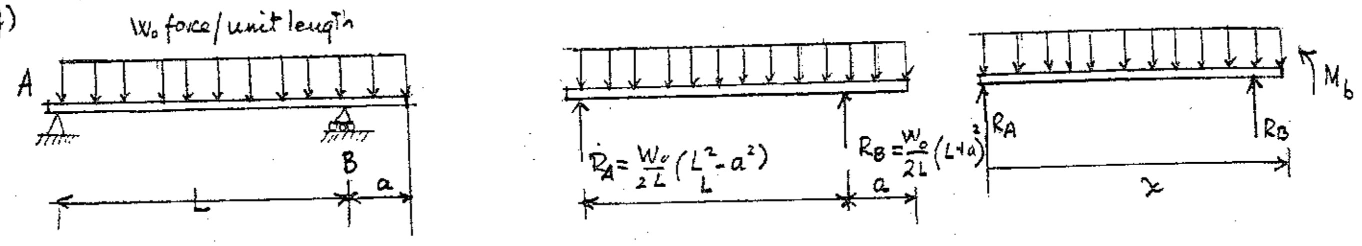
(8-1)(f)
(cont'd)

$$EI\,v = -\frac{W_0L^2x^2}{4} + \frac{W_0Lx^3}{6} - \frac{Wx^4}{24} + C_2$$

$$(v)_{x=0} = 0 \qquad C_2 = 0$$

$$\boxed{v = \frac{W_0L}{2EI}\left[\frac{Lx^2}{2} + \frac{x^3}{3} - \frac{x^4}{12L}\right]}$$

(8-1)(g)



$$R_A = \frac{W_0}{2L}(L^2 - a^2)$$

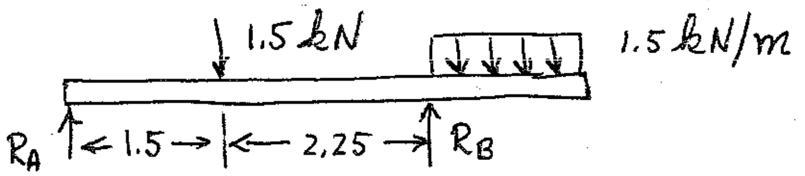
$$R_B = \frac{W_0}{2L}(L+a)^2$$

$$M_b = R_Ax + R_B\langle x-L\rangle^1 - \frac{W_0x^2}{2} = EI\frac{d^2v}{dx^2}$$

$$(v)_{x=0} = 0 \qquad (v)_{x=L} = 0$$

$$\boxed{v = \frac{1}{EI}\left[\frac{R_a}{6}(x^3 - L^2x) - \frac{W_0}{24}(x^4 - L^3x) + \frac{R_B}{6}\langle x-L\rangle^3\right]}$$

(8-1)(h)



FROM ~~PROB~~ 3.8
$R_A = 0.45$ kN
$R_B = 3.3$ kN

$$EI\frac{d^2v}{dx^2} = M = 0.45\langle x\rangle^1 - 1.5\langle x-1.5\rangle^1 + 3.3\langle x-3.75\rangle^1 - 0.75\langle x-3.75\rangle^2$$

$$EI\,v' = \frac{0.45}{2}\langle x\rangle^2 - \frac{1.5}{2}\langle x-1.5\rangle^2 + \frac{3.3}{2}\langle x-3.75\rangle^2 - 0.25\langle x-3.75\rangle^3 + C_1$$

$$EI\,v = \frac{0.45}{6}\langle x\rangle^3 - \frac{1.5}{6}\langle x-1.5\rangle^3 + \frac{3.3}{6}\langle x-3.75\rangle^3 - \frac{0.25}{4}\langle x-3.75\rangle^4 + C_1x + C_2$$

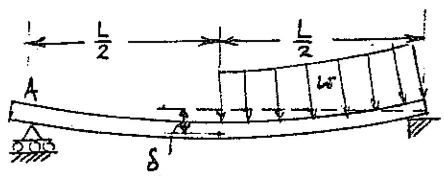
$$v(x=0) = 0 \Rightarrow C_2 = 0$$

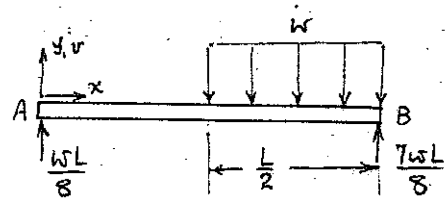
$$v(x=0.375) = 0 \Rightarrow C_1 = 2.689$$

$\therefore$

$$EI\,v = 0.075\langle x\rangle^3 - 0.25\langle x-1.5\rangle^3 + 0.55\langle x-3.75\rangle^3 + 0.0625\langle x-3.75\rangle^4 - 2.689$$

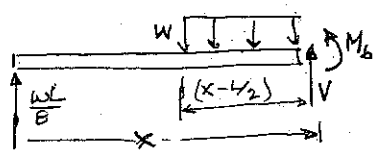
8.2





Bending Moment:

$$EI\frac{d^2v}{dx^2} = M_b = \frac{wL}{8}x - \frac{w}{2}\langle x-\tfrac{L}{2}\rangle^2 .$$



Integrating, $EI\,v = \frac{wLx^3}{48} - \frac{w}{24}\langle x-\tfrac{L}{2}\rangle^4 + C_1 x + C_2$

Boundary Conditions: at $x=0$, $v=0$; at $x=L$, $v=0$

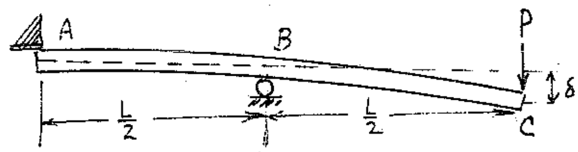
These give $C_2 = 0$, $C_1 = -\frac{7wL^3}{384}$

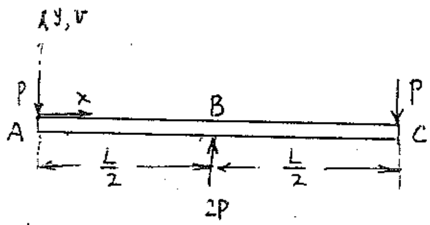
$$\therefore EIv = \frac{wLx}{48}\left(x^2 - \frac{7}{8}L^2\right) - \frac{w}{24}\langle x - \tfrac{L}{2}\rangle^4$$

$$\delta = (-v)_{x=\frac{L}{2}} = \frac{5wL^4}{768EI}$$

8.3

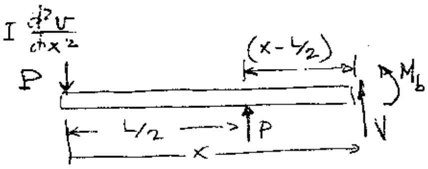
Bending Moments:





$M_b = -Px + 2P\langle x - \tfrac{L}{2}\rangle^1$

$M_b = EI\frac{d^2v}{dx^2}$



$$\therefore\quad EI\frac{d^2v}{dx^2} = -Px + 2P\langle x-\tfrac{L}{2}\rangle^1$$

Integrating twice, $EIv = -\frac{Px^3}{6} + \frac{P}{3}\langle x - \tfrac{L}{2}\rangle^3 + C_1x + C_2$

where C_1, C_2 are constants of integration.

Boundary Conditions: at $x=0$, $v=0$

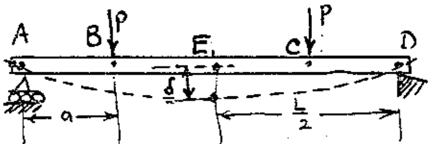
at $x = \frac{L}{2}$, $v = 0$

Using these, $C_2 = 0$, $C_1 = \frac{PL^2}{24}$

$$\therefore\quad EIv = \frac{Px}{6}\left(\frac{L^2}{4} - x^2\right) + \frac{P}{3}\langle x - \tfrac{L}{2}\rangle^3$$

$$\delta = (-v)_{x=L} = \frac{PL^3}{12EI}$$

8.4

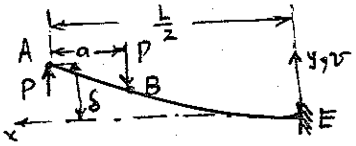


From Symmetry, reactions at A, D are = P.

Method of superposition: From symmetry, slope at $E_1 = 0$.

Consider AE to be a cantilever fixed at E, at zero slope.

Displacement of A, v_a, consists of two parts:



v_a = (v_{a1} due to load at A) + (v_{a2} due to load at B)

v_{a2} = (v_B due to load at B) + $a \times \left[\left(\frac{dv}{dx}\right)_B \text{ due to load at B} \right]$

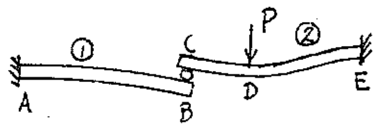
Using Table 8.1, we may write: v_B due to load at B $= -\frac{P(\frac{L}{2}-a)^3}{3EI} = -\frac{P(L-2a)^3}{24EI}$

$\left(\frac{dv}{dx}\right)_B$ due to load at B $= -\frac{P(\frac{L}{2}-a)^2}{2EI} = -\frac{P(L-2a)^2}{8EI}$

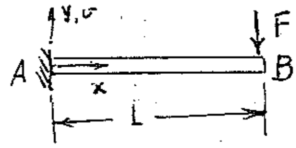
v_{a1} due to load at A $= \frac{P(\frac{L}{2})^3}{3EI} = \frac{PL^3}{24EI}$. Combining all these, and writing $\delta = v_a$,

$$\delta = v_a = -\frac{P}{8EI}\left\{\frac{(L-2a)^3 - L^3}{8} + a\,(L-2a)^2\right\} = \frac{Pa}{24EI}(3L^2 - 4a^2)$$

8.5



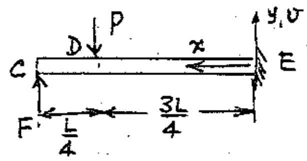
Beam ①:



Geometric Compatibility:

$$v_B = v_c \quad \text{-----} \quad (i)$$

Beam ②:



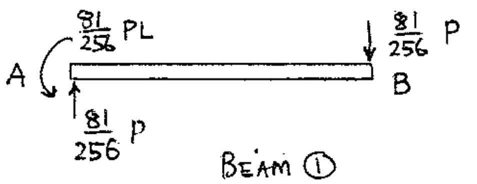
Displacements: Use the method of superposition employed in problem 8.4.

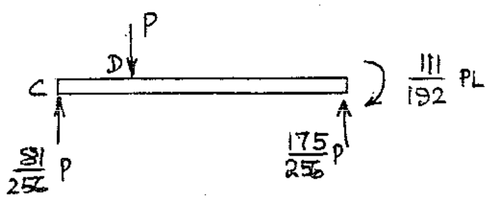
Beam ①: $$v_B = -\frac{FL^3}{3EI}$$

Beam ②: $$v_c = -\left\{ \frac{P\left(\frac{3L}{4}\right)^3}{3EI} + \frac{L}{4}\cdot\frac{P\left(\frac{3L}{4}\right)^2}{2EI} - \frac{FL^3}{3EI} \right\}$$

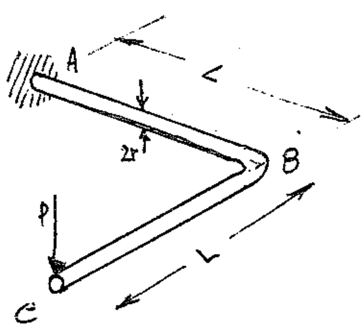
Substituting these in (i), we get $F = \frac{81}{256}P$.

Reactions are shown below:





8.6



Method of superposition:

$$\delta_c = (\delta_c)_{bending} + (\delta_c)_{torsion}$$

$(\delta_c)_{bending} = (\delta_{BA})$ of B relative to A caused by bending-moment due to P + (δ_{CB}) of C relative to B caused by bending-moment due to P

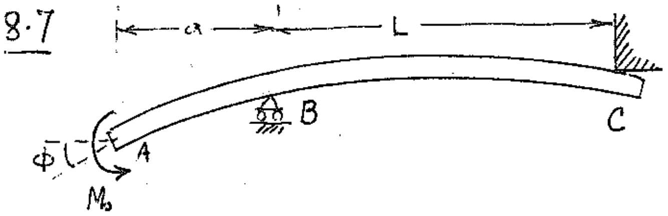
$$(\delta_c)_{torsion} = L \times (\phi_{BA})_{torsion}$$

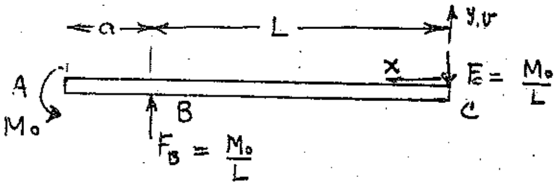
$$\delta_{BA} = \frac{PL^3}{3EI} \; ; \; I = \frac{\pi}{4}r^4 \; \therefore \; \delta_{BA} = \frac{4PL^3}{3\pi E r^4}$$

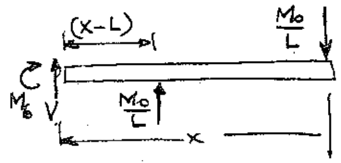
$$\delta_{CB} = \frac{PL^3}{3EI} = \frac{4PL^3}{3\pi E r^4} \; ; \quad \phi_{BA} = \frac{M_t L^2}{G I_z} = \frac{PL\cdot L^2}{G\cdot\frac{\pi}{2}r^4}$$

. Combining all these, $\delta_c = \frac{8PL^3}{3\pi E r^4} + \frac{2PL^3}{\pi G r^4}$; $G = \frac{E}{2(1+\nu)}$. This gives $\delta_c = \frac{4PL^3}{3\pi E r^4}(5+3\nu)$

8.7







Bending Moments: $EI\frac{d^2v}{dx^2} = -\frac{M_o}{L}x + \frac{M_o}{L}\langle x-L\rangle^1$

Integrating, $EI\frac{dv}{dx} = -\frac{M_o}{2L}x^2 + \frac{M_o}{2L}\langle x-L\rangle^2 + C_1$

and $EI\,v = -\frac{M_o}{6L}x^3 + \frac{M_o}{6L}\langle x-L\rangle^3 + C_1x + C_2$

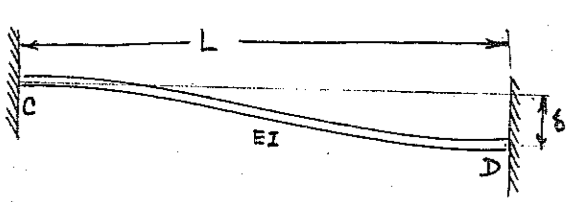
Boundary Conditions: at $x=0,\ v=0$; at $x=L,\ v=0$ } These give $C_2 = 0,\ C_1 = \frac{M_oL}{6}$

$$\therefore \frac{dv}{dx} = \frac{1}{EI}\left(-\frac{M_o}{2L}x^2 + \frac{M_o}{2L}\langle x-L\rangle^2 + \frac{M_oL}{6}\right)$$

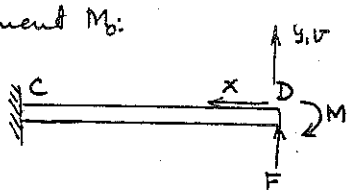
$$\phi = -\left(\frac{dv}{dx}\right)_{x=L+a} = \frac{1}{EI}\left(\frac{M_o}{2L}(L+a)^2 - \frac{M_o}{2L}a^2 - \frac{M_oL}{6}\right) = \frac{M_o(L+3a)}{3EI}$$

8.8

Consider one of the beams, labeled CD. All the beams deform in the same manner.



Let the reaction at D consist of a force F and a moment M_o:

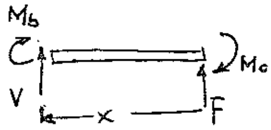


Bending Moment:

$$EI\frac{d^2v}{dx^2} = M_b = Fx - M_o$$

Integrating, $EI\frac{dv}{dx} = F\frac{x^2}{2} - M_ox + C_1$

$$EI\,v = F\frac{x^3}{6} - M_o\frac{x^2}{2} + C_1x + C_2$$



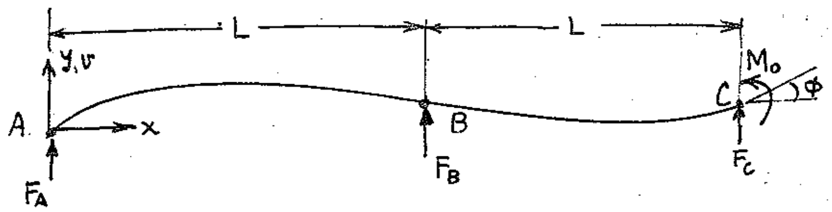
Boundary conditions: at $x=0,\ v=0,\ \frac{dv}{dx}=0$

at $x=L,\ v=\delta,\ \frac{dv}{dx}=0.$

These give $C_1 = C_2 = 0,\quad F = \frac{12EI\delta}{L^3},\quad M_o = \frac{6EI\delta}{L^2}$

$\therefore M_b = \frac{12EI\delta}{L^3}x - \frac{6EI\delta}{L^2}.\quad |M_b|_{max}$ at $x=0,\ x=L = \frac{6EI\delta}{L^2}$

8.9



Equilibrium: $\sum F_y = 0$: $F_A + F_B + F_C = 0$ ------- (1)

$\sum M_{Bz} = 0$: $F_A - F_C = \frac{M_0}{L}$ ------ (2)

Bending Moment: $EI\frac{d^2v}{dx^2} = M_b = F_A x + F_B \langle x-L\rangle^1$

Integrating, $EI\frac{dv}{dx} = F_A\frac{x^2}{2} + \frac{F_B}{2}\langle x-L\rangle^2 + C_1$

$EI\,v = F_A\frac{x^3}{6} + \frac{F_B}{6}\langle x-L\rangle^3 + C_1x + C_2$

Boundary conditions: at $x=0$, $v=0$

$x=L$, $v=0$

$x=2L$, $v=0$

Thus we have two equations (1) & (2) and three boundary conditions, and five unknowns, F_A, F_B, F_C, C_1 and C_2 : the problem has a unique solution.

Get: $C_2 = 0$, $C_1 = -\frac{F_A}{6}L^2$

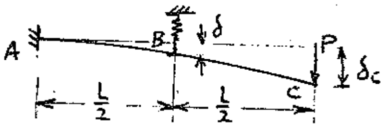
$$\frac{8F_A}{6} + \frac{F_B}{6} - \frac{2F_A}{6} = 0 \quad \text{-------} \quad (3)$$

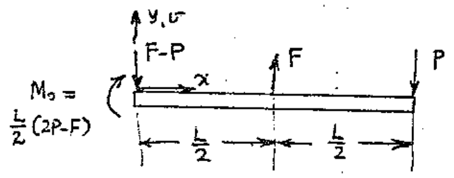
from the boundary conditions. From (1), (2) and (3),

$F_A = -\frac{M_0}{4L}$; $F_B = \frac{3M_0}{2L}$; $F_C = -\frac{5M_0}{4L}$.

Then $\phi \cong \left(\frac{dv}{dx}\right)_{x=2L} = \frac{7M_0L}{24EI}$

8.10





Bending Moments: $EI\dfrac{d^2v}{dx^2} = M_b = (P-F)x + \frac{L}{2}(2P-F) + F\langle x - \frac{L}{2}\rangle^1$

Integrating, $EI\dfrac{dv}{dx} = (P-F)\dfrac{x^2}{2} + \dfrac{L}{2}(2P-F)x + \dfrac{F}{2}\langle x-\frac{L}{2}\rangle^2 + C_1$

$$EIv = (P-F)\frac{x^3}{6} + \frac{L}{4}(2P-F)x^2 + \frac{F}{6}\langle x-\tfrac{L}{2}\rangle^3 + C_1x + C_2$$

Boundary Conditions: at $x=0$, $v=0$, $\dfrac{dv}{dx}=0$.

These give $C_1 = C_2 = 0$

Force-Displacement for spring is $F = -kv_B = -k(v)_{x=\frac{L}{2}}$

or $F = -\dfrac{k}{EI}\left\{\dfrac{7PL^3}{48} - \dfrac{FL^3}{12}\right\}$ $\therefore$ $F = \dfrac{7P}{4}\dfrac{\chi}{\chi-1}$ where $\chi \equiv \dfrac{kL^3}{12EI}$

$\delta = -v_B = \dfrac{F}{k} = \dfrac{7P}{4k}\left\{\dfrac{kL^3/12EI}{kL^3/12EI - 1}\right\}$ (not required)

$\delta_c = (-v)_{x=L}$ may be thrown, after some algebra, into the following form:

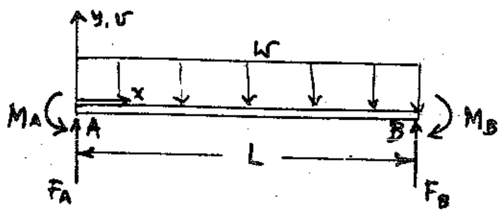
$$\delta_c = \frac{PL^3}{3EI}\left[1 - \frac{25}{32\left(\frac{24EI}{kL^3}+1\right)}\right]$$

8.11 Equilibrium: $\sum F_y = 0$: $F_A + F_B = wL$

From Symmetry, $M_A = M_B$

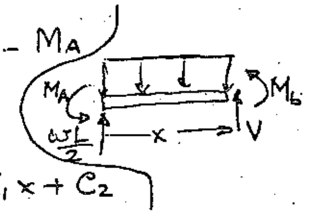
$F_A = F_B = \dfrac{wL}{2}$

Then, $\sum M_{Az} = 0$: Identically satisfied



8·11 continued

Bending Moments: $EI\dfrac{d^2v}{dx^2} = M_b = wL\dfrac{x}{2} - w\dfrac{x^2}{2} - M_A$



Integrating, $EI\dfrac{dv}{dx} = wL\dfrac{x^2}{4} - w\dfrac{x^3}{6} - M_A x + C_1$

$$EIv = wL\frac{x^3}{12} - w\frac{x^4}{24} - M_A\frac{x^2}{2} + C_1 x + C_2$$

Boundary Conditions: at $x=0$, $v = \dfrac{dv}{dx} = 0$ ---- (i)

at $x=L$, $v = \dfrac{dv}{dx} = 0.$ ---- (ii)

From (i), $C_1 = C_2 = 0$.

From either of (ii) (the other is contained in the "symmetry" argument),

$$M_A = \frac{wL^2}{12}$$

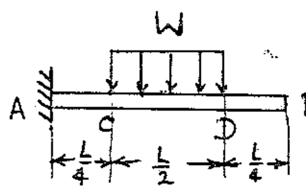
$\therefore M_b = wL\dfrac{x}{2} - \dfrac{wL^2}{12} - w\dfrac{x^2}{2}$

$|M_b|_{max}$ at $\dfrac{dM_b}{dx} = 0$, ie., at $x = \dfrac{L}{2}$

$$|M_b|_{max} = \frac{wL^2}{24}.$$

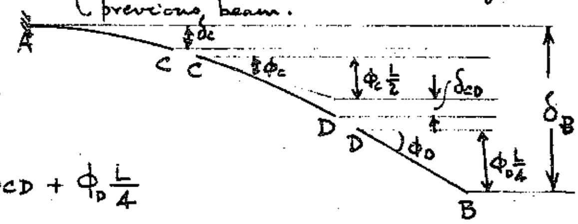
$|v|_{max}$ occurs at $x = \frac{L}{2}$, $\quad |v|_{max} = \dfrac{wL^4}{384EI}$

8·12



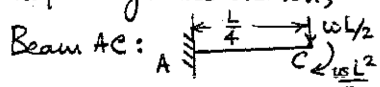
Define $w \equiv \dfrac{2W}{L}$

Superposition: { Assume the beam consists of three cantilever beams built in at the end slope of the previous beam.



Referring to the sketch, $\delta_B = \delta_C + \phi_C\dfrac{L}{2} + \delta_{CD} + \phi_D\dfrac{L}{4}$

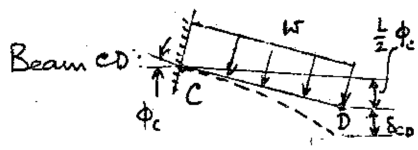
Beam AC:



$$\delta_C = \frac{\left(\frac{wL}{2}\right)\left(\frac{L}{4}\right)^3}{3EI} + \frac{\left(\frac{wL^2}{8}\right)\left(\frac{L}{4}\right)^2}{2EI} = \frac{5wL^4}{768EI}$$

$$\phi_C = \frac{\left(\frac{wL}{2}\right)\left(\frac{L}{4}\right)^2}{2EI} + \frac{\left(\frac{wL^2}{8}\right)\left(\frac{L}{4}\right)}{EI} = \frac{3wL^3}{64EI}$$

Beam CD:

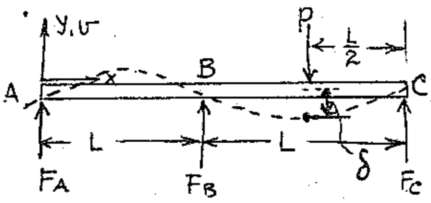


$$\delta_{CD} = \frac{w\left(\frac{L}{2}\right)^4}{8EI} = \frac{wL^4}{128EI}; \quad \phi_{CD} = \frac{w\left(\frac{L}{2}\right)^3}{6EI} = \frac{wL^3}{48EI}$$

$$\therefore \phi_D = \phi_C + \phi_{CD} = \frac{13wL^3}{192EI}.$$

Combining all these, $\delta_B = \dfrac{5wL^4}{768EI} + \dfrac{3wL^3}{64EI}\cdot\dfrac{L}{2} + \dfrac{wL^4}{128EI} + \dfrac{13wL^3}{192EI}\cdot\dfrac{L}{4} = \dfrac{7wL^4}{128EI} = \dfrac{7}{64}\dfrac{WL^3}{EI}$

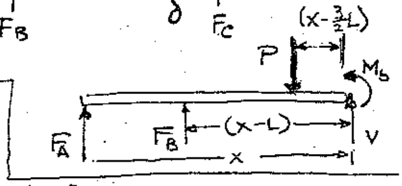
8.13 Equilibrium:



$\sum F_y = 0: \quad F_A + F_B + F_C = P \quad \text{--- (1)}$

$\sum M_{Bz} = 0: \quad F_C - F_A = \frac{P}{2} \quad \text{--- (2)}$

Bending Moment:



$$EI\frac{d^2v}{dx^2} = M_b = F_A x + F_B \langle x-L\rangle^1 - P\langle x - \tfrac{3L}{2}\rangle$$

Integrating, $EIv = F_A\frac{x^3}{6} + \frac{F_B}{6}\langle x-L\rangle^3 - \frac{P}{6}\langle x - \frac{3L}{2}\rangle^3 + C_1x + C_2$

Boundary Conditions, $v = 0$ at $x = 0, L, 2L$.

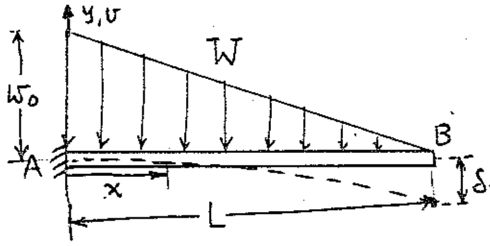
Using these conditions and equations (1) and (2),

$$C_2 = 0; \quad C_1 = \frac{PL^2}{64}; \quad F_A = -\frac{3}{32}P; \quad F_B = \frac{22}{32}P; \quad F_C = \frac{13}{32}P.$$

$$\therefore \quad v = \frac{1}{EI}\left\{-\frac{Px^3}{64} + \frac{11P}{96}\langle x-L\rangle^3 - \frac{P}{6}\langle x - \frac{3L}{2}\rangle^3 + \frac{PL^2}{64}x\right\}$$

$$\delta = -(v)_{x=\frac{3L}{2}} = \frac{23PL^3}{1536EI}$$

8.14 Let w_0 = rate of loading at A.



$$w_0 \cdot \frac{1}{2}\cdot L = W \quad \therefore \quad w_0 = \frac{2W}{L}$$

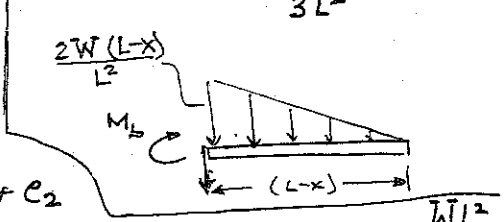
At x, the rate of loading is

$$\frac{L-x}{L}w_0 = \frac{2W(L-x)}{L^2}$$

Bending Moment at x due to loading beyond x is $M_b = -\frac{2W(L-x)}{L^2}\cdot\frac{(L-x)}{2}\cdot\frac{(L-x)}{3}$

$$= -\frac{W(L-x)^3}{3L^2}$$

$$EI\frac{d^2v}{dx^2} = M_b = -\frac{W(L-x)^3}{3L^2}$$



Integrating, $EI\frac{dv}{dx} = \frac{W(L-x)^4}{12L^2} + C_1$

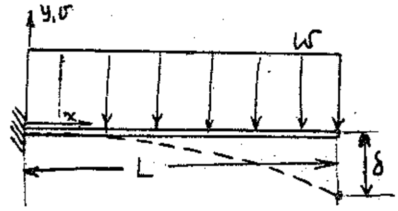
$$EIv = -\frac{W(L-x)^5}{60L^2} + C_1x + C_2$$

Boundary Conditions: $v = \frac{dv}{dx} = 0$ at $x=0$. These give $\begin{cases} C_1 = -\frac{WL^2}{12} \\ C_2 = \frac{WL^3}{60}\end{cases}$

$$v = \frac{1}{EI}\left\{-\frac{W(L-x)^5}{60L^2} - \frac{WL^2x}{12} + \frac{WL^3}{60}\right\}. \quad \delta_B = -(v)_{x=L} = \frac{WL^3}{15EI}$$

8.15

(a) Bending Moments:

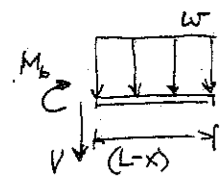


$$EI\frac{d^2v}{dx^2} = M_b = -\frac{w(L-x)^2}{2}$$

$$EI\frac{dv}{dx} = \frac{w(L-x)^3}{6} + C_1$$

$$EI\,v = -\frac{w(L-x)^4}{24} + C_1x + C_2$$

Boundary Conditions : at $x=0,\ v = \frac{dv}{dx} = 0.$



These give $C_1 = -\frac{wL^3}{6},\quad C_2 = \frac{wL^4}{24}$

$$\therefore\ v = \frac{1}{EI}\left\{-\frac{w(L-x)^4}{24} - \frac{wL^3x}{6} + \frac{wL^4}{24}\right\}$$

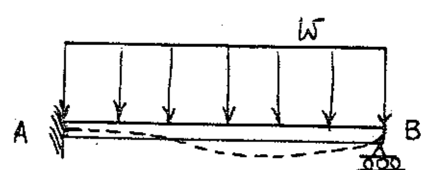
$|v|_{max} = \delta$ occurs at $x = L$. $\quad |v|_{max} = \frac{wL^4}{8EI}$

at $x=0$, $\quad |M_b|_{max} = \frac{wL^2}{2}$

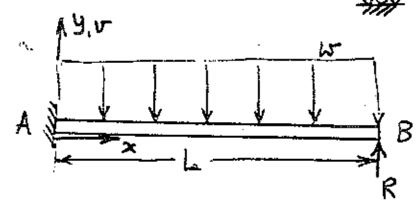
(b)

Method of superposition:

The reaction R is such that the deflection at $B = 0$.



Deflection at B due to $R = \frac{RL^3}{3EI}$



Deflection at B due to $w = -\frac{wL^4}{8EI}$

$$\therefore\ \frac{RL^3}{3EI} = \frac{wL^4}{8EI}, \quad \text{or} \quad R = \frac{3}{8}wL$$

Bending Moments: $M_b = EI\frac{d^2v}{dx^2} = \frac{3}{8}wL(L-x) - \frac{w}{2}(L-x)^2$

$$\therefore\ EI\frac{dv}{dx} = -\frac{3}{16}wL(L-x)^2 + \frac{w}{6}(L-x)^3 + C_1$$

$$EI\,v = +\frac{3}{48}wL(L-x)^3 - \frac{w}{24}(L-x)^4 + C_1x + C_2$$

Boundary Conditions. $v = \frac{dv}{dx} = 0$ at $x=0$.

These give, after solving for C_1 and C_2:

$$v = \frac{1}{EI}\left\{\frac{1}{16}wL(L-x)^3 - \frac{w(L-x)^4}{24} + \frac{wL^3x}{48} - \frac{wL^4}{48}\right\}$$

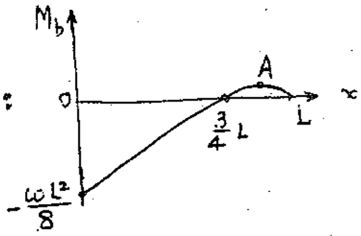
$|v|_{max}$ occurs at $\frac{dv}{dx} = 0$, $\quad$ i.e., $-\frac{3}{16}wL(L-x)^2 + \frac{w(L-x)^3}{6} + \frac{wL^3}{48} = 0$

8.15 continued

An approximate solution is $x = 0.58L$

$$\therefore \quad |v|_{max} = 0.42\,\frac{wL^4}{48EI}$$

The bending-moment diagram is :



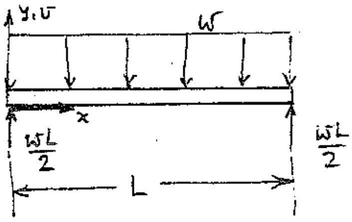
An extremum occurs at A, where

$\frac{dM_b}{dx} = 0$. This is at $x = \frac{5}{8}L$.

$$(M_b)_A = \frac{9}{128}wL^2 \ . \text{ But } (M)_{x=0} = -\frac{wL^2}{8}$$

$$\therefore \ |(M_b)_{x=0}| > |(M_b)_A| \ . \quad \therefore \ |(M_b)_{max}| = \frac{wL^2}{8}, \text{ at } x=0.$$

8.16 (a) Bending Moment:



$$EI\frac{d^2v}{dx^2} = \frac{wL}{2}x - \frac{wx^2}{2}$$

$$\therefore \ EIv = \frac{wL}{12}x^3 - \frac{wx^4}{24} + C_1x + C_2$$

Boundary Conditions: $v=0$ at $x=0$, $x=L$.

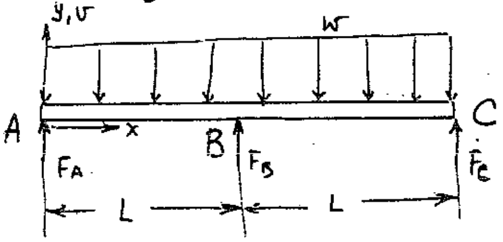
These give $C_2 = 0$, $C_1 = -\frac{wL^3}{24}$.

$$\therefore \quad v = \frac{1}{EI}\left\{\frac{wLx^3}{12} - \frac{wx^4}{24} - \frac{wL^3x}{24}\right\}.$$

$|v|_{max}$ and $|M_b|_{max}$ occur at $x = \frac{L}{2}$

$$|v|_{max} = \frac{5wL^4}{384EI} \qquad |M_b|_{max} = \frac{wL^2}{8}.$$

(b) Equilibrium requirements give



$$F_A = F_C \quad \text{---- (i)}$$

$$F_A + F_B + F_C = 2wL \quad \text{---- (ii)}$$

Bending Moment:

$$EI\frac{d^2v}{dx^2} = M_b = F_A \cdot x + F_B\langle x-L\rangle^1 - \frac{wx^2}{2}$$

$$\therefore \ EIv = \frac{F_A}{6}x^3 + \frac{F_B}{6}\langle x-L\rangle^3 - \frac{wx^4}{24} + C_1x + C_2$$

Boundary Conditions: $v=0$ at $x=0$, $x=L$, $x=2L$.

Using these and (i) and (ii), get $C_2 = 0$, $C_1 = -\frac{wL^3}{48}$,

8.16 continued

and $F_A = F_C = \dfrac{3wL}{8}$, $F_B = \dfrac{5}{4}wL$

Then $v = \dfrac{1}{EI}\left\{ \dfrac{3Lx^3}{48} + \dfrac{10}{48}L\langle x-L\rangle^3 - \dfrac{x^4}{24} - \dfrac{xL^3}{48} \right\}$

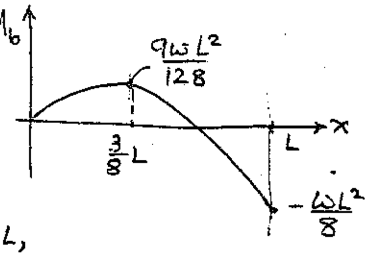
$|v|_{max}$ occurs at $x \approx 0.45L$, where $\dfrac{dv}{dx} = 0$.

$$\underline{\underline{|v|_{max}}} = \frac{wL^4}{200EI} = \underline{\underline{1.91\,\frac{wL^4}{384EI}}}$$

For AB, $M_b = \dfrac{3wLx}{8} - \dfrac{wx^2}{2}$

An extremum occurs at $x = \frac{3}{8}L$, where

$(M_b)_{x=\frac{3}{8}L} = \dfrac{9wL^2}{128}$.



However, the largest value occurs at $x = L$,

even though it is not an extremum: $\underline{\underline{|M_b|_{max} = \dfrac{wL^2}{8}}}$.

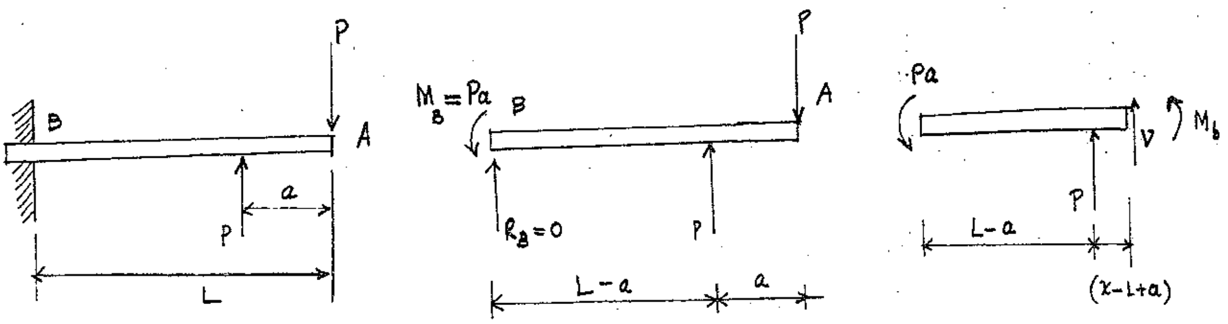
Thus: In going from case (a) to case (b), we

(i) increase central support force from wL to $\frac{5}{4}wL$.

(ii) don't change max. bending moment; change its location.

(iii) decrease max. deflection from $5\dfrac{wL^4}{384EI}$ to $1.91\dfrac{wL^4}{384EI}$.

(8-17) (a)



$\Sigma F_y = 0$

$R_B = P - P = 0$

$\Sigma M = 0$

$-M_B + PL - P(L-a) = 0$

$M_B = Pa$

$$M_b = -Pa + P\langle x-L+a\rangle^1 = EI\frac{d^2v}{dx^2}$$

$$\frac{dv}{dx} = \frac{-Pa}{EI}x + \frac{P\langle x-L+a\rangle^2}{2EI} + C_1$$

$$\left(\frac{dv}{dx}\right)_{x=0} = 0 \rightarrow C_1 = 0$$

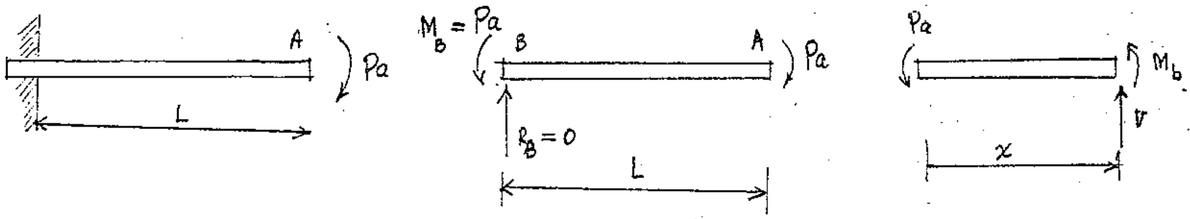
$$v = -\frac{Pax^2}{2EI} + \frac{P\langle x-L+a\rangle^3}{6EI} + C_2$$

$$(v)_{x=0} = 0 \rightarrow C_2 = 0$$

$$\boxed{v = -\frac{Pax^2}{2EI} + \frac{P\langle x-L+a\rangle^3}{6EI}}$$

$$(v)_{x=L} = \frac{PaL^2}{2EI}\left[\frac{a^2}{3L^2} - 1\right]$$

(8-17) cont
(b)



$\Sigma F_y = 0$ $\qquad\qquad$ $\Sigma M = 0$

$R_B = 0$ $\qquad\qquad$ $M_B = Pa$

$$M_b = -Pa = EI\frac{d^2v}{dx^2}$$

$$\frac{dv}{dx} = -\frac{Pa}{EI}x + C_1$$

$$\left(\frac{dv}{dx}\right)_{x=0} = 0 \rightarrow C_1 = 0$$

$$v = -\frac{Pa\,x^2}{2EI} + C_2$$

$$(v)_{x=0} = 0 \rightarrow C_2 = 0$$

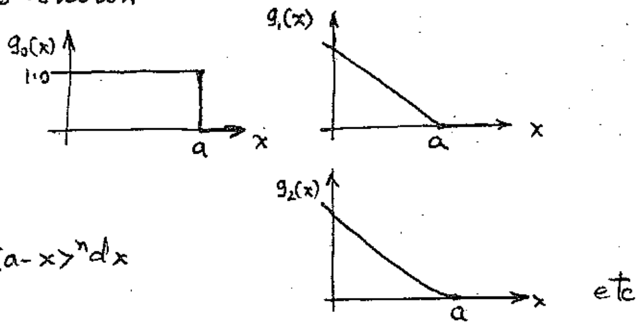
$$\boxed{v = -\frac{Pa\,x^2}{2EI}}$$

$$(v)_{x=L} = -\frac{PaL^2}{2EI}$$

for (a)

$$\lim_{\frac{a}{L}\to 0} (v)_{x=L} = -\frac{PaL^2}{2EI}$$ this equals to $(v)_{x=L}$ for (b)

<u>8.18</u> The functions appear as shown



$$g_n(x) = \langle a-x \rangle^n$$

ie., $$g_n(x) = \begin{cases} (a-x)^n & x \leq a \\ 0 & x > a \end{cases}$$

Now $$\int_x^\infty \langle a-x\rangle^n dx = \int_x^a \langle a-x\rangle^n dx + \int_a^\infty \langle a-x\rangle^n dx$$

From the definitions, $\int_a^\infty \langle a-x\rangle^n dx = 0$

Also, $$\int_x^a \langle a-x\rangle^n dx = \begin{cases} \dfrac{(a-x)^{n+1}}{n+1} & x \leq a \\ 0 & x > a \end{cases}$$

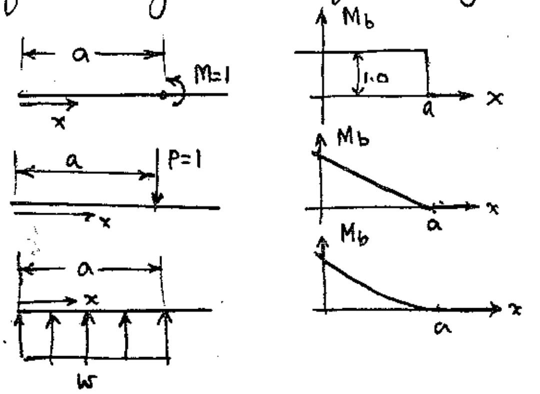
This proves that $$\int_x^\infty \langle a-x\rangle^n dx = \frac{\langle a-x\rangle^{n+1}}{(n+1)}$$

What $\int \langle a-x\rangle^n dx$ signifies is the integral between an arbitrary fixed point x_0 and a variable point x.

Let $x_0 < a$. $$\int_{x_0}^x \langle a-x\rangle^n dx = \begin{cases} \dfrac{(a-x_0)^{n+1}}{n+1} - \dfrac{(a-x)^{n+1}}{n+1} = C - \dfrac{(a-x)^{n+1}}{n+1} & x \leq a \\ \dfrac{(a-x_0)^{n+1}}{n+1} = C & x > a \end{cases}$$

$$\therefore \int \langle a-x\rangle^n dx \equiv \int_{x_0}^x \langle a-x\rangle^n dx = -\frac{\langle a-x\rangle^{n+1}}{n+1} + C.$$

Consider the following loads on beams and the bending moments in the beams caused solely by these loads, as we proceed from right to left along the beam.



Comparing these M_b sketches with those for $g_n(x)$, we see that $g_n(x)$ can represent the bending moments due to these loadings.

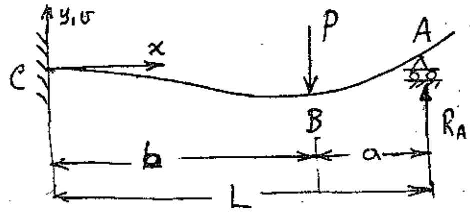
$g_0(x) \rightarrow$ moment due to a unit couple

$g_1(x) \rightarrow$ moment due to a unit point load

$g_2(x) \rightarrow$ moment due to a unit continuous load, etc.

8.19

Writing $M_b = \sum_{j=1}^{n} P_j \langle a_j - x \rangle^1$,



We have $n = 2$,

$P_1 = -P \qquad a_1 = b$

$P_2 = R_A \qquad a_2 = L$

Also, $a_2 > x$ for all x in this problem, and so $\langle a_2 - x \rangle^1 = (a_2 - x) = (L - x)$

$$\therefore \quad M_b(x) = R_A (L-x) - P\langle b-x\rangle^1 = EI\frac{d^2v}{dx^2}$$

and $EI\dfrac{dv}{dx} = -\dfrac{R_A}{2}(L-x)^2 + \dfrac{P}{2}\langle b-x\rangle^2 + C_1$

$$EI\,v = \frac{R_A}{6}(L-x)^3 - \frac{P}{6}\langle b-x\rangle^3 + C_1 x + C_2$$

Boundary Conditions: at $x=0$, $v = \dfrac{dv}{dx} = 0$; at $x = L$, $v=0$.

These give $C_1 = \frac{1}{2}(R_A L^2 - P b^2)$

$$C_2 = -\frac{1}{6}(R_A L^3 - P b^3)$$

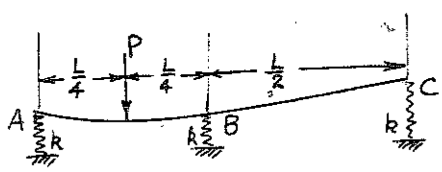
and $R_A = \dfrac{P b^2}{2 L^3}(3L - b)$

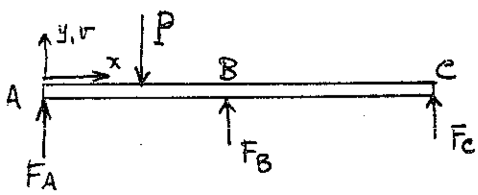
8.20 Fix the origin at A in the deflected position. Then, if

δ = displacement of A, from

geometry $\delta - v_B$ = displacement of B

$\delta - v_C$ = displacement of C.





Force - deflection: $F_A = k\delta$

$$F_B = k(\delta - v_B) = F_A - k v_B$$

$$F_C = k(\delta - v_C) = F_A - k v_C$$

8.20 continued

Equilibrium: $\sum F_y = 0$: $F_A + F_B + F_C = P$, or $3F_A - k(v_B + v_C) = P$ ----- (1)

$$\sum M_{A_z} = 0: \quad \frac{PL}{4} = F_B \frac{L}{2} + F_C L = L\left\{ \frac{3F_A}{2} - k\left(\frac{v_B}{2} + \frac{v_C}{2}\right)\right\}$$

or $6F_A - 2k(v_B + v_C) = P$ ------- (2)

Bending Moment:

$$EI\frac{d^2v}{dx^2} = M_b = F_A \cdot x - P\langle x - \tfrac{L}{4}\rangle^1 + F_B\langle x - \tfrac{L}{2}\rangle^1$$

$$= F_A \cdot x - P\langle x - \tfrac{L}{4}\rangle^1 + (F_A - kv_B)\langle x - \tfrac{L}{2}\rangle^1$$

Integrating, $EIv = \frac{F_A}{6}x^3 - \frac{P}{6}\langle x - \frac{L}{4}\rangle^3 + \frac{(F_A - kv_B)}{6}\langle x - \frac{L}{2}\rangle^3 + C_1 x + C_2$

Boundary Conditions: at $x = 0$, $v = 0$

$x = \frac{L}{2}$, $v = v_B$

$x = L$, $v = v_C$

These give: $C_2 = 0$,

$$EIv_B = \frac{F_A L^3}{48} - \frac{PL^3}{384} + C_1\frac{L}{2} \qquad \text{---------- (3)}$$

$$EIv_C = \frac{F_A L^3}{6} - \frac{27PL^3}{384} + \frac{(F_A - kv_B)L^3}{48} + C_1 L \qquad \text{------ (4)}$$

From (1), (2), (3) and (4), obtain, after considerable algebra:

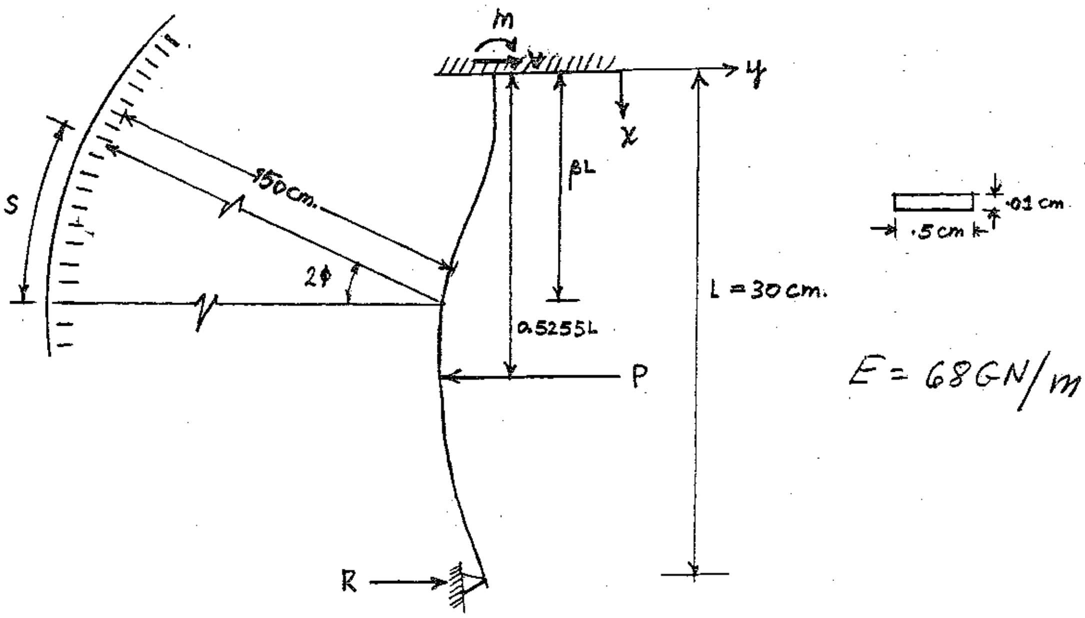
$$F_A = \frac{P\left(\frac{7}{2} + \frac{13}{384}N\right)}{\left(6 + \frac{N}{12}\right)}$$

where $N \equiv \dfrac{kL^3}{EI}$

$$F_B = \frac{P\left(2 + \frac{22}{384}N\right)}{\left(6 + \frac{N}{12}\right)}$$

$$F_C = \frac{P\left(\frac{1}{2} - \frac{3}{384}N\right)}{\left(6 + \frac{N}{12}\right)}$$

1)



$E = 68 GN/m^2$

(a) Max. deflection at max. slope $\longrightarrow EI \frac{d^2y}{dx^2} = 0$

$$EI \frac{d^2y}{dx^2} = M + Vx - P\langle x - 0.5255L\rangle^1 \qquad B.C. (1)\ x = 0 \quad y = 0$$
$$(2)\ x = 0 \quad \frac{dy}{dx} = 0$$

$$EI \frac{dy}{dx} = Mx + \frac{Vx^2}{2} - \frac{P}{2}\langle x - 0.5255L\rangle^2 + C_1 \qquad C_1 = 0 \text{ from } B.C.(2)$$

$$EI\, y = \frac{Mx^2}{2} + \frac{Vx^3}{6} - \frac{P}{6}\langle x - 0.5255L\rangle^3 + C_2 \qquad C_2 = 0 \text{ from } B.C.(1)$$

B.C. $x = L = 30$ cm, $y = \frac{d^2y}{dx^2} = 0$

$\therefore$ (1) $M + VL - P(0.4745L) = 0$

(2) $\frac{ML^2}{2} + V\frac{L^3}{6} - \frac{P}{6}(0.4745L)^3 = 0$

Eliminating M from (1) + (2), $V\left(\frac{L^3}{6} - \frac{L^3}{2}\right) + P\left[\frac{0.4745}{2} - \frac{(0.4745)^3}{6}\right]L^3 = 0$

$$\therefore V = 3P\left[\frac{0.4745}{2} - \frac{(0.4745)^3}{6}\right] = 0.658P$$

Eliminating V from (1) + (2), $M\left(\frac{L^2}{2} - \frac{L^2}{6}\right) + P\left[\frac{0.4745}{6} - \frac{(0.4745)^3}{6}\right]L^3 = 0$

$$\therefore M = -3P\left[\frac{0.4745 - (0.4745)^3}{6}\right]L = -0.184LP$$

For max. deflection $M + Vx = 0 \quad \therefore x = -\frac{M}{V} = \frac{0.184}{0.658}L = 0.279L$

$$\therefore \beta = 0.279$$

8-21 con't

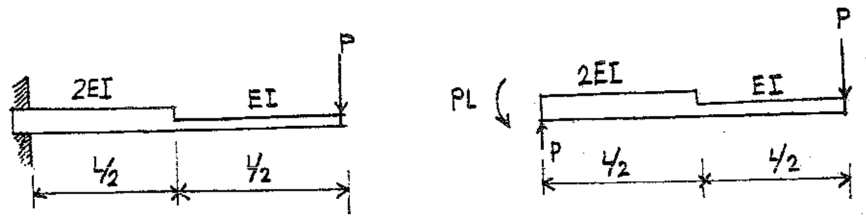
(b) $$EI\frac{dy}{dx} = Mx + \frac{Vx^2}{2} \; ; \quad \phi_{max} = \left.\frac{dy}{dx}\right|_{max} = \frac{1}{EI}\left[(-0.184)(0.279L)PL + \frac{0.658(0.279)^2L^2}{2}\right]$$

$$I = \frac{bh^3}{12} = \frac{(0.5)(0.01)^3}{12} = 4.2\times10^{-8}\ cm^4 \; ; \qquad E = 68\ GN/m^2$$

$$\phi_{max} = \frac{-0.051+0.024}{EI}PL^2 = 8.7\times10^{-4}\,P$$

(c) $$k = \frac{S}{P} = \frac{(2\phi_{max})R}{P} = \frac{2\times8.7\times10^{-4}\,P\times150}{P} = 0.261\ cm/dyne$$

(8.22)



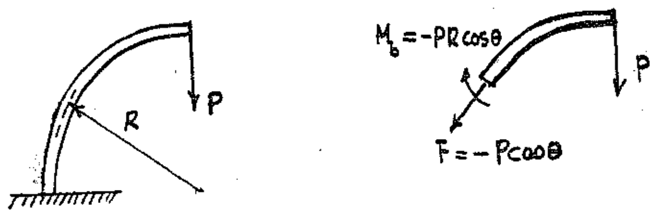
$$M_b = P(x-L) \quad\longrightarrow\quad U = \int_0^{L/2}\frac{M_b^2}{4EI}dx + \int_{L/2}^{L}\frac{M_b^2}{2EI}dx$$

$$= \frac{P^2}{4EI}\left[\frac{x^3}{3} - Lx^2 + L^2x\right]_0^{L/2} + \frac{P^2}{2EI}\left[\frac{x^3}{3} - Lx^2 + L^2x\right]_{L/2}^{L}$$

$$= \frac{9P^2L^3}{96EI}$$

$$\boxed{\delta = \frac{\partial U}{\partial P} = \frac{3PL^3}{16EI}}$$

(8.23)



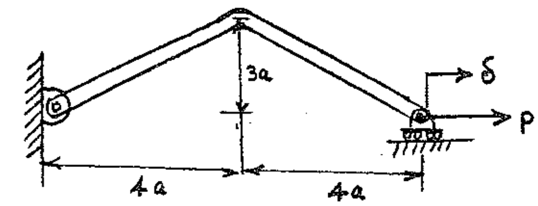
AXIAL FORCE $F = -P\cos\theta$

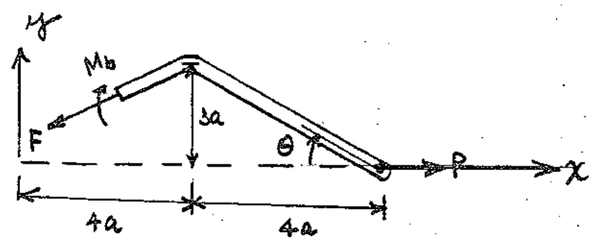
bending moment $M_b = -PR\cos\theta$

$$U = \int_L \frac{F^2}{2AE}dx + \int_L \frac{M_b^2}{2EI}dx = \frac{P^2R}{2AE}\int_0^{\pi/2}\cos^2\theta\, d\theta + \frac{P^2R^3}{2EI}\int_0^{\pi/2}\cos^2\theta\, d\theta$$

$$\therefore \frac{\text{contribution due to axial force}}{\text{contribution due to bending}} = \frac{PR}{AE}\cdot\frac{EI}{PR^3} = \frac{R}{\pi r^2E}\cdot\frac{E\pi r^4}{4R^3} = \frac{1}{4}\frac{r^2}{R^2}$$

-24)





$0 < x \leq 4a \qquad M_{b1} = Px\tan\theta = \frac{3xP}{4}$

$\qquad\qquad\qquad\quad F = P\cos\theta = \frac{4}{5}P$

$4a < x \leq 8a \qquad M_{b2} = P(8a-x)\tan\theta = \frac{3P}{4}(8a-x)$

$\qquad\qquad\qquad\quad F = P\cos\theta = \frac{4P}{5}$

$$U = \int \frac{M_b^2}{2EI}dx + \int \frac{F^2}{2AE}dx$$

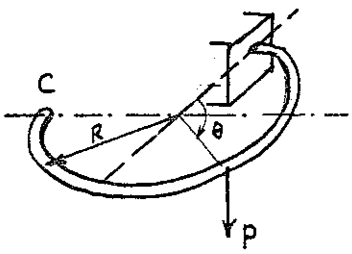
$$= \frac{1}{2EI}\int_0^{4a}\left(\frac{3xP}{4}\right)^2 dx + \frac{1}{2EI}\int_{4a}^{8a}\left[\frac{3P}{4}(8a-x)\right]^2 dx + \frac{1}{2AE}\int_0^{8a}\left(\frac{4P}{5}\right)^2 dx$$

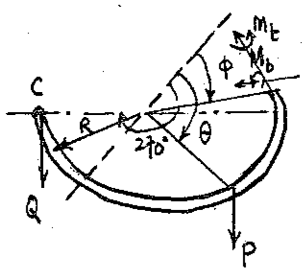
$$= \frac{9P^2}{32EI}\left[\frac{x^3}{3}\right]_0^{4a} + \frac{9P^2}{32EI}\left[64a^2x - 8ax^2 + \frac{x^3}{3}\right]_{4a}^{8a} + \frac{8}{25}\frac{P^2}{AE}\left[x\right]_0^{8a}$$

$$= \frac{12P^2a^3}{EI} + \frac{64P^2a}{25AE}$$

$$\therefore\ \delta = \frac{\partial U}{\partial P} = \frac{24Pa^3}{EI} + \frac{128Pa}{25AE}$$

(8.25)





Use energy method.
Add fictitious load Q at C.
M_b = bending moment
M_t = twisting moment

M_b due to load P $= PR\sin(\theta-\phi)$
M_t " " " " $= PR[1-\cos(\theta-\phi)]$ (see Prob. 3.27)

M_b including load Q $= R[P\sin(\theta-\phi) - Q\cos\phi]$
M_t " " " $= R\{P[1-\cos(\theta-\phi)] + Q(1+\sin\phi)\}$

$$U = \underbrace{\int_L \frac{M_t^2}{2GI}dx}_{I} + \underbrace{\int_L \frac{M_b^2}{2EI}dx}_{II}$$

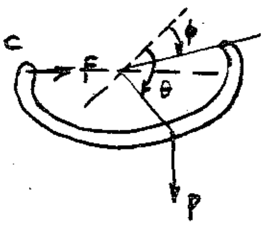
$$I = \frac{R^3}{2GI}\int_0^{3\pi/2}\left\{P^2[1-\cos(\theta-\phi)]^2 + 2PQ(1+\sin\phi)[1-\cos(\theta-\phi)] + Q^2(1+\sin\phi)^2\right\}d\phi$$

$$II = \frac{R^3}{2EI}\int_0^{3\pi/2}\left\{P^2\sin(\theta-\phi) - 2PQ\cos\phi\sin(\theta-\phi) + Q^2\cos^2\phi\right\}d\phi$$

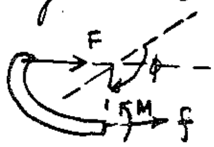
$$\lim_{Q\to 0}\frac{\partial I}{\partial Q} = \frac{R^3P}{GI}\left[\frac{3}{2}\pi - 1 - \sin\theta\left(1+\frac{3}{4}\pi\right) - \frac{3}{2}\cos\theta\right] \; ; \; \lim_{Q\to 0}\frac{\partial II}{\partial Q} = -\frac{PR^3}{EI}\left[\frac{3\pi}{4}\sin\theta - \frac{1}{2}\cos\theta\right]$$

$$\boxed{\therefore\ \delta_c = \frac{\partial U}{\partial Q} = \frac{R^3P}{GI}\left[\frac{3\pi}{2} - 1 - \left\{\sin\theta\left(1+\frac{3}{4}\pi\right)\right\} - \frac{3}{2}\cos\theta\right] - \frac{PR^3}{EI}\left[\frac{3}{4}\pi\sin\theta - \frac{1}{2}\cos\theta\right]}$$ vertical deflection

For the horizontal deflection, add a horizontal fictitious force F at C.



consider only the horizontal force F



$M_b = -FR\cos\phi$
$f = -F\cos\phi$
(axial force)

$$U = \int \frac{M_t^2}{2GI}dx + \int \frac{M_b^2}{2EI}dx + \int \frac{F^2}{2AE}dx$$

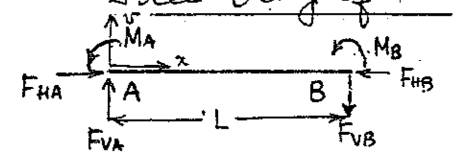
$$\frac{\partial U}{\partial F} = \frac{\partial}{\partial F}\int \frac{(PR\sin(\theta-\phi) - FR\cos\phi)^2}{2EI}R\,d\phi + \frac{\partial}{\partial F}\int \frac{F^2\cos^2\phi}{2AE}R\,d\phi$$

$$\lim_{F\to 0}\frac{\partial U}{\partial F} = \frac{PR^3}{2EI}\left\{\cos\theta - \frac{3\pi}{2}\sin\theta\right\}$$

horizontal deflection
$$\delta_h = \lim_{F\to 0}\frac{\partial U}{\partial F} = \frac{PR^3}{2EI}\left\{\cos\theta - \frac{3\pi}{2}\sin\theta\right\}$$

8.26 We shall consider this problem in detail, as an illustrative case.

Free-body of AB:



Equilibrium:

$$F_{HB} = F_{HA} \quad \text{------} \quad (1)$$

$$F_{VA} - F_{VB} = 0 \quad \text{------} \quad (2)$$

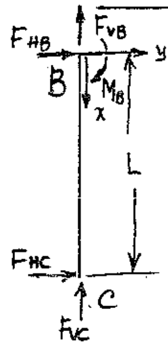
$$F_{VA} = \frac{M_A + M_B}{L} \quad \text{------} \quad (3)$$

Bending Moment:

$$EI\frac{d^2v}{dx^2} = M_b = F_{VA}\cdot x - M_A \quad \text{------} \quad (A)$$

Boundary Conditions: at $x=0$, $v = \frac{dv}{dx} = 0$ ------ (4,5)

Free-body of BC



Equilibrium:

$$F_{VB} = -F_{VC} \quad \text{------} \quad (6)$$

$$F_{HB} + F_{HC} = 0 \quad \text{------} \quad (7)$$

$$F_{HB} = \frac{M_B}{L} \quad \text{------} \quad (8)$$

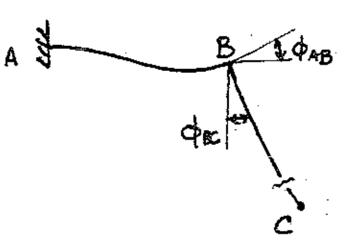
Bending Moment:

$$EI\frac{d^2v}{dx^2} = M_b = F_{HB}\cdot x + M_b \quad \text{------} \quad (B)$$

Boundary Conditions: $v=0$ at $x=0$ ------ (9)

$v = \delta$ at $x = L$ ------ (10)

Geometric Compatibility: $\phi_{AB} = \phi_{BC}$ (see sketch) ------ (11)



This means weld remains right angled.

If extension of BC is negligible (which it is), B moves downwards.

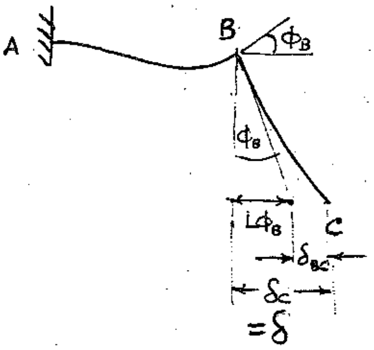
Unknowns: M_A, F_{VA}, F_{VB}, M_B, F_{VB}, F_{HB}, F_{VC}, F_{HC} and 4 constants of integration associated with (A) & (B) = 12 unknowns.

Equations: 12 (numbered): problem has a unique solution.

We will make a negligible error in writing $(v_B)_{AB} = 0$.

Once these formal arguments are over, the method of superposition.

may be used to advantage



Beam AB: $\delta_B = 0 = -\dfrac{M_B L^2}{2EI} + \dfrac{F_{VB} L^3}{3EI}$

$\therefore \quad M_B = \frac{2}{3} F_{VB} L$

$$\phi_B = \frac{M_B L}{EI} - \frac{F_{VB} L^2}{2EI} = \frac{F_{VB} L^2}{6EI}$$

Beam BC: $\delta_C = \delta = L\phi_B + \delta_{BC}$

$\delta = 25\,mm$

$$\delta_{BC} = \frac{F_{HC} L^3}{3EI} \quad \therefore \quad \delta_C = \frac{F_{VB} L^3}{6EI} - \frac{F_{HC} L^3}{3EI}$$

We have $M_B = \frac{2}{3} F_{VB} L$ and, from (7) and (8), $M_B = L F_{HB} = -L F_{HC}$

Substituting for F_{VB} and F_{HC},

$$\delta_C = \frac{L^3}{EI}\left(\frac{M_B}{3L} + \frac{M_B}{4L}\right) = \frac{7 M_B L^2}{12 EI} \quad \therefore \quad \underline{\underline{M_B = \frac{12 EI \delta_C}{7 L^2}}}$$

Also $F_{VB} = \dfrac{18}{7}\dfrac{EI\delta_C}{L^3}$, $\quad M_A = L F_{VA} - M_B = L F_{VB} - M_B = \dfrac{6}{7}\dfrac{EI\delta_C}{L^2}$

$\therefore \quad |M_A| < |M_B|$.

Max stress occurs at surface of rod, due to combined bending and axial stresses. Shear stresses are negligible. Both axial and bending stresses are larger at B than at A. Max stress occurs at B.

$$\text{Bending stress} = \sigma_2 = +\frac{M_b\left(\frac{d}{2}\right)}{I} = \frac{12}{7}\frac{\delta_C E}{L^2}\frac{d}{2}$$

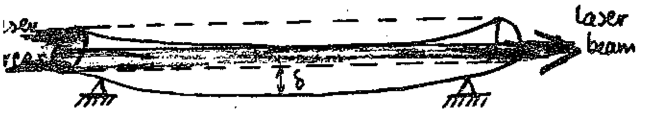
$$= \frac{12}{7}\,\frac{25\times10^{-3} \times 205\times10^{3} \times 13\times10^{-3}}{(1.25)^2 \times 2} = 36.5\,MN/m^2$$

$$\text{AXIAL STRESS} = \sigma_1 = \frac{F_{HB}}{\frac{\pi}{4}d^2} = \frac{4}{\pi\,(13)^2}\,\frac{12}{7}\,\frac{EI\delta_C}{L^3}$$

$$= 2.37\times10^{-2}\,MN/m^2$$

$\therefore$ MAXIMUM STRESS $= 36.5\,MN/m^2$

3.27)



Cross-section

$A_2 = \frac{1}{2}A$ neglect effect of rotation of end

we will treat Tube as a uniformly loaded beam under its own weight (from Table 8.1)

$$\delta = \frac{5}{384}\frac{WL^4}{EI} \qquad W = \gamma A \ N/m; \ I = A\rho^2 \longrightarrow \boxed{L^4 = \frac{384}{5}\frac{E}{\gamma}\rho^2\delta} \cdots\cdots (1)$$

AREA OF INTERSECTION OF TWO CIRCLES

$$\frac{A}{\pi r^2} = \frac{2}{\pi}\sin^{-1}\left(\frac{d}{r}\right) - \frac{2d}{\pi r}\left(1 - \frac{\delta}{2r}\right) \quad ; \ d^2 = \delta\left[r - \frac{\delta}{4}\right]$$

TYPICAL VALUES

(δ/r)	$A/\pi r^2$
0	0
0.5	0.144
1.0	0.391
1.6	0.747
2.0	1.00

FOR $A/\pi r^2 = \frac{1}{2}$, $\frac{\delta}{r} = 1.2$; $\delta = 75 \times 1.2 = 90$ mm

$$L^4 = \frac{384}{5}\frac{E}{\gamma}\rho^2\delta$$

$$\gamma = 76.44 \frac{kN}{m^2}$$

$$I_{yy} \approx A\left(\frac{r_o^2 + r_i^2}{4}\right) = A\rho^2$$

WHEN $t = 3mm$; $\rho^2 = 2.927 \times 10^{-3} m^2$

$$\therefore L^4 = \frac{384}{5}\frac{205\times10^9 \times 2.927\times10^{-3} \times 90\times10^{-3}}{76.44\times10^3}$$

$$L = 15.26 m \quad *$$

INCREASE WALL THICKNESS to 12mm, $r_o = 87$, $r_i = 75$

$$\rho^2 = 3.298 \times 10^{-3}$$

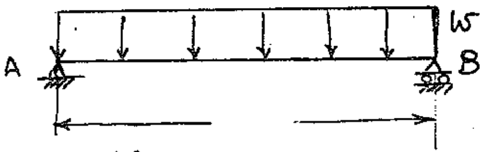
$$\delta = 90 \times \frac{3.298}{2.927} = 101.4 \text{ mm}$$

$$\therefore \frac{\delta}{r} = 1.35$$

$$\therefore \frac{A}{\pi r^2} = 0.595$$

NOTE That INCREASE IN THICKNESS, GIVES INCREASE IN δ

8.28



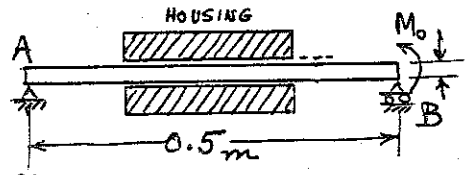
$w = 15\ kN/m$
$E = 7 GN/m^2$
$L = 3.6 m$

Using the result of problem 8.16 (a),

$$\delta_{max} = \frac{5wL^4}{384EI} \quad \text{or} \quad \frac{\delta_{max}}{L} = \frac{5wL^3}{384EI} = \frac{1}{360}$$

This gives $I = \frac{5 \times 360}{384} \times \frac{15 \times (3.6)^3}{7 \times 10 6} = 4.68 \times 10^{-4}\ m^4$

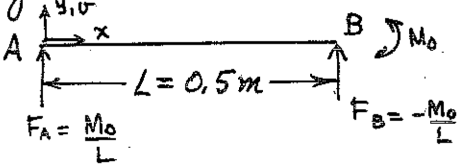
8.29



$d = 12mm$

Since the distortion of the cross-section is negligible, the clearance becomes 2.5mm when the shaft centre-line moves 0.10" in any direction at any point within the housing.

In the free-body alongside, the weight of the shaft is neglected.



Bending Moment:

$$EI\frac{d^2v}{dx^2} = M_b = \frac{M_o}{L}x$$

$$\therefore EIv = \frac{M_o}{6L}x^3 + C_1x + C_2.$$

Boundary Conditions: $v=0$ at $x=0$, $x=L$.

These give $C_2 = 0$, $C_1 = -\frac{M_oL}{6}$

$$\therefore v = \frac{1}{EI}\left\{\frac{M_o}{6}\left(\frac{x^3}{L} - xL\right)\right\}$$

$|v|_{max}$ occurs at $x = \frac{L}{\sqrt{3}}$, where $\frac{dv}{dx} = 0$.

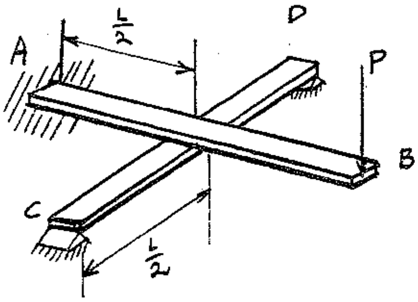
$|v|_{max} = \frac{M_oL^2}{9\sqrt{3}EI} = 2.5$, required. $\therefore M_o = \frac{2.5 \times 10^{-3} \times 9\sqrt{3}\,EI}{L^2}$

$I = \frac{\pi}{64} \times (\frac{1}{2})^2\ in^4$; $\therefore M_o = 32.5\ N \cdot m$

$= 1.018 \times 10^{-9} m^4$

$E = 205\ GN/m^2$

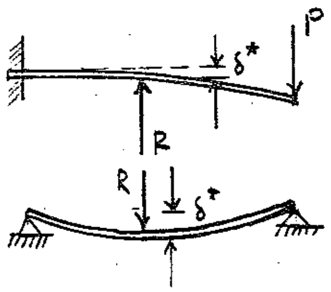
(8-30)



Consider superposition of cases shown

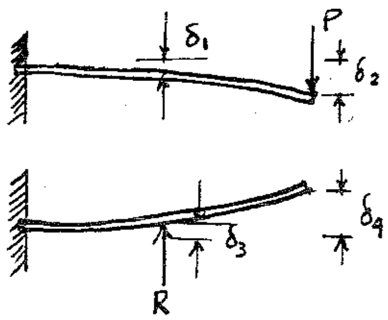
(a)

(b)



to find R such that δ^* in both cases are equal

Case (a) is the superposition of two cases



From Table 8.1.

$$\delta_1 = \frac{5}{48}\frac{PL^3}{EI} \quad ; \quad \delta_2 = \frac{PL^3}{3EI}$$

$$\delta_3 = \frac{1}{24}\frac{RL^3}{EI} \qquad \delta_4 = \frac{5}{48}\frac{RL^3}{EI}$$

$$\therefore \quad \delta^* = \delta_1 - \delta_3 = \frac{5}{48}\frac{PL^3}{EI} - \frac{1}{24}\frac{RL^3}{EI}$$

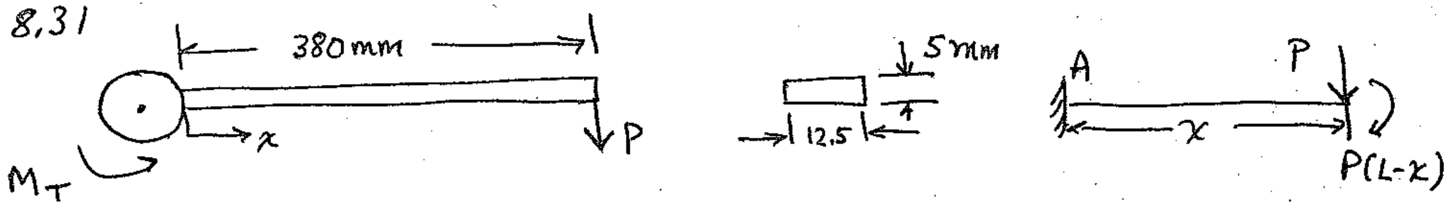
From Table 8.1, δ^* for case (b) is $\frac{1}{48}\frac{RL^3}{EI}$

$$\therefore \quad \frac{5}{48}\frac{PL^3}{EI} - \frac{1}{24}\frac{RL^3}{EI} = \frac{1}{48}\frac{RL^3}{EI} \longrightarrow R = \frac{5}{3}P$$

Deflection under load P is $= \delta_2 - \delta_4 = \frac{PL^3}{3EI} - \frac{5}{48}\cdot\frac{5}{3}\frac{PL^3}{EI}$

$$\delta = \frac{23}{144}\frac{PL^3}{EI}$$

8.31

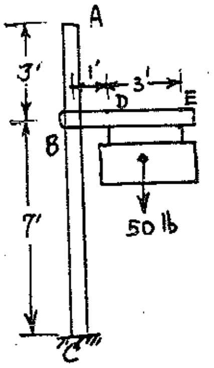


EQUILIBRIUM $P = \frac{M_T}{L}$ CONSIDER DEFLECTION AT x by SUPERPOSITION

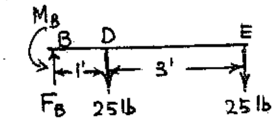
$$\delta_x = \frac{Px^3}{3EI} + \frac{P(L-x)x^2}{2EI} = \frac{P}{EI}\left[\frac{x^3}{3} + \frac{x^2(L-x)}{2}\right]$$

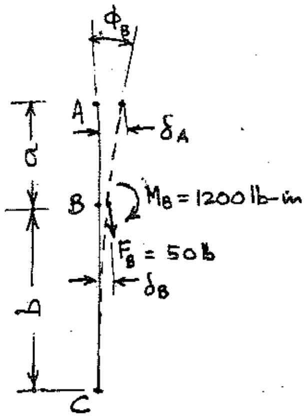
at $x = 300$, $\delta_x = 1.24 \times 10^{-3} M_T$ $(M_T : N.m)$

8.32



Free-body of BE:





Equilibrium: $F_B = 50$ lb

$M_B = 1200$ lb-in

Note that the weight of the signpost and the force F_B contribute to the bending-moment in the signpost, but the effect is negligible.

Method of superposition (see sketch): $\delta_A = \delta_B + AB \times \phi_B$

$= \delta_B + a\phi_B$

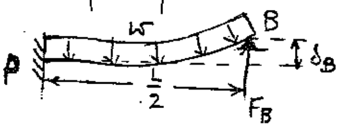
$$\delta_B = \frac{M_B b^2}{2EI}, \quad \phi_B = \frac{M_B b}{EI}, \quad a = 36'', \quad b = 84''$$

$$\therefore \delta_A = \frac{bM_B}{EI}\left(\frac{b}{2} + a\right) = \frac{84 \times 1200}{\frac{\pi}{64}(256 - 150) \times 30 \times 10^6} \times 78 = \underline{\underline{0.0503''}}$$

8.33 If the pontoon sinks by an amount δ_P, the upthrust is

$$F_P = \gamma A \delta_P .$$

By symmetry, the reactions at each end are equal. Also by symmetry, the bridge has zero slope at its centre. Consider each half to be a cantilever fixed at the centre at zero slope and use the method of superposition:

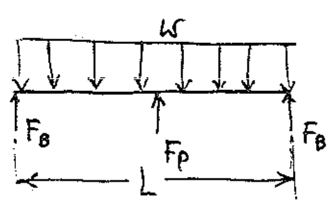


$$\delta_B = \frac{F_B \left(\frac{L}{2}\right)^3}{3EI} - \frac{w\left(\frac{L}{2}\right)^4}{8EI}$$

$$= \frac{L^3}{8EI}\left(\frac{F_B}{3} - \frac{wL}{16}\right).$$

This is also δ_P relative to B, when B is considered fixed.

$$\therefore \quad F_P = \gamma A \delta_P = \frac{\gamma A L^3}{8EI}\left(\frac{F_B}{3} - \frac{wL}{16}\right)$$



For Equilibrium of the whole bridge

$$2F_B + F_P = wL.$$

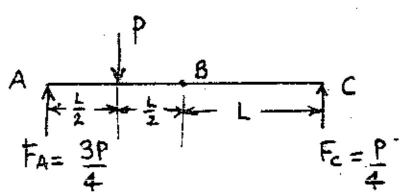
This gives $F_B = \dfrac{wL\left(1 + \dfrac{\gamma A L^3}{128EI}\right)}{\left(2 + \dfrac{\gamma A L^3}{128EI}\right)}$ $\quad \therefore \quad \delta_P = \dfrac{5wL^4}{384EI\left(1 + \dfrac{\gamma A L^3}{48EI}\right)}$

8.34 Method of superposition.

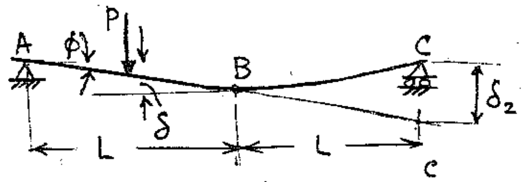
Let slope of rigid beam $= \phi$

Then (see sketch) $\delta \approx \phi L$

$$\delta_2 \approx 2\phi L$$



Consider BC as a cantilever fixed at B, at slope ϕ:

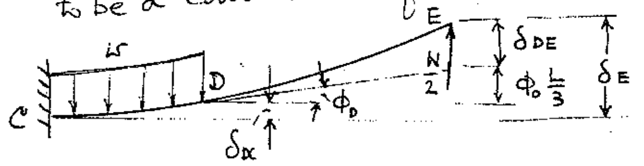


$$\delta_2 = \frac{P}{4}\frac{(L)^3}{3EI} = 2\phi L$$

$$\therefore \quad \phi = \frac{PL^2}{24EI}$$

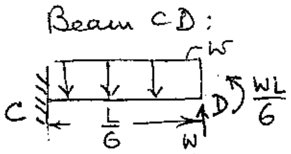
$$(\delta)_{L/2} = \phi\frac{L}{2} = \frac{PL^3}{48EI}$$ = deflection under load.

8.35 Approximate the box by a continuous load, $w = \frac{3W}{L}$.
The reactions at the ends are $\frac{W}{2}$, by symmetry. Also from symmetry, the slope at the centre is zero. Consider each half to be a cantilever fixed at the centre at zero slope.



$$\delta_E = \delta_{DC} + \phi_D \frac{L}{3} + \delta_{DE}$$

Beam CD:



$$\delta_{DC} = -\frac{w\left(\frac{L}{6}\right)^4}{8EI} + \frac{W}{2}\frac{\left(\frac{L}{6}\right)^3}{3EI} + \frac{WL}{6}\frac{\left(\frac{L}{6}\right)^2}{2EI} = \frac{29\,WL^3}{10{,}368\,EI}$$

$$\phi_D = -\frac{w\left(\frac{L}{6}\right)^3}{6EI} + \frac{W}{2}\frac{\left(\frac{L}{6}\right)^2}{2EI} + \frac{WL}{6}\frac{\left(\frac{L}{6}\right)}{EI} = \frac{42\,WL^2}{1296\,EI}$$

$$\delta_{DE} = \frac{\frac{W}{2}\left(\frac{L}{3}\right)^3}{3EI} = \frac{WL^3}{162\,EI}$$

$$\therefore \delta_E = \frac{WL^3}{EI}\left(\frac{1}{162} + \frac{29}{10368} + \frac{14}{1296}\right) = \frac{205\,WL^3}{10{,}368\,EI}$$

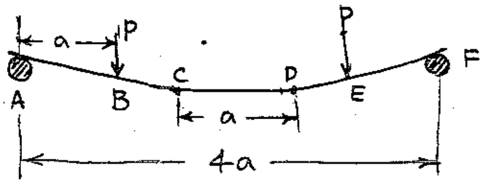
This is also δ_c relative to a fixed E: $\delta_c = \frac{205\,WL^3}{10{,}368\,EI}$

Approximate solution: replace box by a concentrated force at centre.
From Table 8.1, p 378

$$\delta = \frac{PL^3}{48EI}$$

$$\frac{\delta_{approx}}{\delta_{exact}} = \frac{10{,}368}{48 \times 205} = 1.05.$$ There is a 5% error in making the approx.

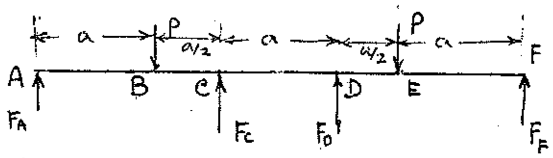
8.36 Reason as in Example 8.4:



(i) $\frac{d^2v}{dx^2} = \frac{M_b}{EI} = 0$ in CD
(ii) $\frac{dM_b}{dx} = V_y$ (shear force) $= 0$ in CD
(iii) Net loading at a point $= 0$ in CD
(iv) Point forces must act at C, D, for otherwise $M_b \neq 0$ in CD.

CONTINUED

8.36 continued



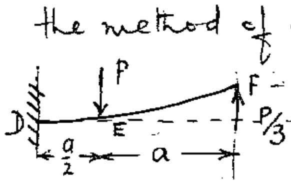
Equilibrium requires

$$F_A + F_C = F_D + F_F = P$$

Constraint: $M_b = 0$ at C and D. This gives $\begin{cases} F_A = F_F = P/3 \\ F_D = F_C = 2P/3 \end{cases}$

With these, $V_y = 0$, $M_b = 0$ in CD.

Consider DF to be a cantilever fixed at D at zero slope, and use the method of superposition:



$$\delta_F = \overbrace{\frac{\frac{Pa}{3}\left(\frac{a}{2}\right)^2}{2EI} - \frac{\left(\frac{2P}{3}\right)\left(\frac{a}{2}\right)^3}{3EI}}^{\text{(DEFLECTION AT E)}} + \overbrace{a\left[\frac{\left(\frac{Pa}{3}\right)\left(\frac{a}{2}\right)}{EI} - \frac{\left(\frac{2P}{3}\right)\left(\frac{a}{2}\right)^2}{2EI}\right]}^{\text{(SLOPE AT E)}} + \overbrace{\frac{\left(\frac{P}{3}\right)(a)^3}{3EI}}^{\text{(DEFLECTION OF EF)}}$$

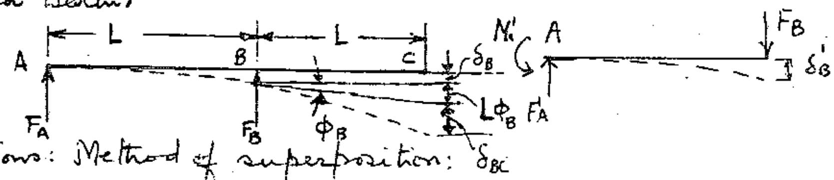
$$= \frac{5Pa^3}{24EI}$$

This is also the deflection of CD relative to a fixed F. Therefore the required condition is

$$\frac{5Pa^3}{24EI} = d, \quad \text{or} \quad P = \frac{24EId}{5a^3}$$

8.37 The following free-bodies may be drawn, with the assumption given in the problem.

Upper Beam; Lower Beam.



Deflections: Method of superposition:

Upper: $\delta_B = \frac{(P - F_B)L^3}{3EI} + \frac{PL\cdot L^2}{2EI}$ $\qquad \delta_B' = \frac{F_B L^3}{3EI}$

$$= \frac{L^3}{6EI}(5P - 2F_B)$$

Geometric Compatibility: $\delta_B = \delta_B'$. $\therefore \frac{F_B L^3}{3EI} = \frac{L^3}{6EI}(5P - 2F_B)$ or $F_B = \frac{5}{4}P$

Then $\delta_C = \delta_B + L\phi_B + \delta_{BC}$ (see sketch)

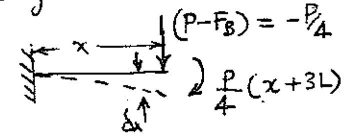
$$= \frac{5}{12}\frac{PL^3}{EI} + L\cdot\left[\frac{PL\cdot L}{EI} + \frac{(-P/4)L^2}{2EI}\right] + \frac{PL^3}{EI} = \frac{13PL^3}{8EI}$$

CONTINUED

8-37 continued

Deflection Curves: method of superposition: consider free-bodies of length x

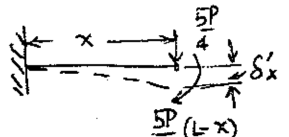
upper:



$x \leq L$

$$\delta_x = \frac{P}{4}\frac{(x+3L)x^2}{2EI} - \frac{P}{4}\frac{x^3}{3EI} = \frac{P}{4EI}\left\{\frac{x^3}{6} + \frac{3}{2}x^2L\right\} \qquad x \leq L$$

Lower:



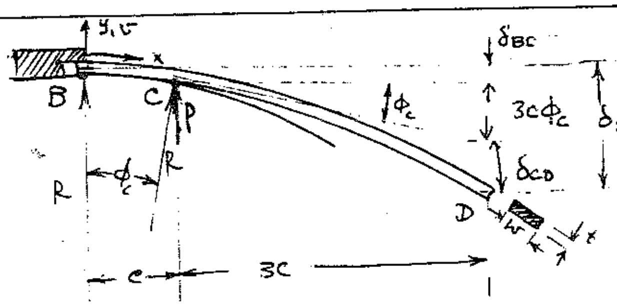
$$\delta'_x = \frac{\left(\frac{5P}{4}\right)x^3}{3EI} + \frac{\left(\frac{5P}{4}\right)(L-x)x^2}{2EI}$$

$$= \frac{P}{4EI}\left\{\frac{5}{2}Lx^2 - \frac{5}{6}x^3\right\} \qquad 0 \leq x \leq L$$

$$\delta'_x - \delta_x = \frac{P}{4EI}(Lx^2 - x^3) > 0 \quad \text{for } 0 < x < L$$
$$= 0 \quad \text{for } x = L$$

∴ The beams touch only at B. The assumption is correct.

8-38



$t \leq R$
$c \leq R$

Reasoning: Over BC

(i) $\frac{d^2v}{dx^2} \approx -\frac{1}{R} = \text{constant} = \frac{M_b}{EI}$

(ii) $\frac{dM_b}{dx} = V_y = 0$ (iii) Intensity of loading is zero

(iv) at C, an upward force P acts, as shown, so that M_b = Constant.

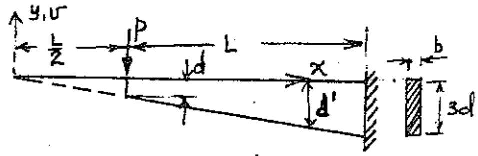
for $x \leq c$, $M_b = -P(4c-x) + P(c-x) = -3Pc \qquad \therefore P = -\frac{M_b}{3c} = \frac{EI}{3cR}$

Superposition: $\delta_D = \delta_{BC} + 3c\phi_c + \delta_{CD}$

$$= \underbrace{\frac{1}{2}\frac{c^2}{R} + 3c \cdot \frac{c}{R}}_{\text{From geometry of circular arc}} + \frac{P(3c)^3}{3EI} = \frac{13c^2}{2R}$$

$$\delta_D = \frac{13}{2}\frac{c^2}{R}$$

8.39



At any x, d' (depth) $= \frac{2dx}{L}$

Then $I_x = \frac{1}{12} b d'^3 = \frac{2bd^3x^3}{3L^3}$

Bending Moments: $EI_x \frac{d^2v}{dx^2} = M_b = -P(x - \frac{L}{2}) \qquad \frac{L}{2} \le x \le \frac{3L}{2}$

Substituting for I_x, $\frac{2Ebd^3}{3L^3}\frac{d^2v}{dx^2} = -\frac{P}{x^3}(x - \frac{L}{2})$

Integrating: $\frac{2Ebd^3}{3L^2}\frac{dv}{dx} = \frac{P}{x} - \frac{PL}{4x^2} + C_1$

$\frac{2Ebd^3}{3L^2} v = P\ln x + \frac{PL}{4x} + C_1 x + C_2$

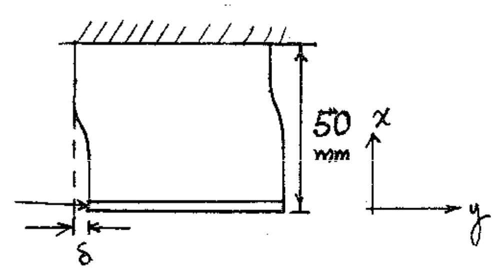
Boundary Conditions: $v = \frac{dv}{dx} = 0$ at $x = \frac{3L}{2}$

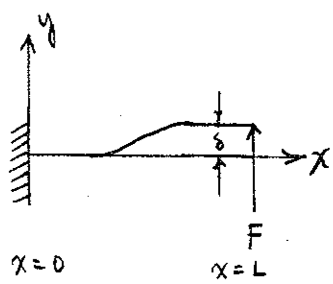
This gives $v = \frac{3PL^3}{2Ebd^3}\left\{ \ln\frac{2x}{3L} + \frac{2}{3} + \frac{L}{4x} - \frac{5}{9}\frac{x}{L} \right\}$

at $x = \frac{L}{2}$, $v = \frac{3PL^3}{2Ebd^3}\left(\frac{8}{9} - \log_e 3\right)$

and $\delta \equiv |v|_{L/2} = \frac{3PL^3}{2Ebd^3}\left(\log_e 3 - \frac{8}{9}\right)$

8-40





$\frac{d^2}{dx^2}\left(EI\frac{d^2v}{dx^2}\right) = q = 0$

For constant EI

$\frac{d^4v}{dx^4} = 0$

$v = \frac{Ax^3}{6} + \frac{Bx^2}{2} + Cx + D$

A, B, C, D are constants

B.C. $x = 0$: $v = 0$ ①, $v' = 0$ ②; $x = L$: $v = \delta$ ③, $v' = 0$ ④

from ① + ②: $D = 0$, $C = 0$,

From ③: $\frac{AL^3}{6} + \frac{BL^2}{2} = \delta$

From ④: $\frac{AL^2}{2} + BL = 0$

$A = \frac{-12\delta}{L^3}$; $B = \frac{6\delta}{L^2}$

$\therefore v = -\frac{2\delta}{L^3}x^3 + \frac{3\delta}{L^2}x^2$

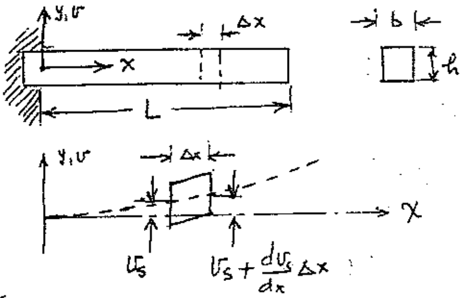
at $x = L$, $V = -F = EI\frac{d^3v}{dx^3} = EI\left(-\frac{12\delta}{L^3}\right) \rightarrow \delta = \frac{FL^3}{12EI}$

a) $v = \frac{F}{2EI}\left[-\frac{x^3}{3} + \frac{Lx^2}{2}\right]$

$M = EI\frac{d^2v}{dx^2} = EI\left\{-\frac{12\delta}{L^3}x + \frac{6\delta}{L^2}\right\}$ $\therefore$ at $x = 0$ $M = M_{max} = \frac{6\delta EI}{L^2}$

) $\sigma_y = -\frac{M_{max} y_{max}}{I_{yy}} = -\frac{6\delta EI}{L^2}\cdot\frac{y_{max}}{I}$ $\therefore \delta_{max} = -\frac{\sigma_y L^2}{6E y_{max}} = \frac{350\times10^6 \times 25\times10^{-4}}{6\times100\times10^9\times0.75\times10^{-3}} = 1.94\,mm$

8.41



$$\frac{dv_s}{dx} = \frac{(\tau_{xy})_{max}}{G} = \frac{3P}{2Gbh}$$

$$\left[(\tau_{xy})_{max} = \frac{3P}{2bh} \text{ for rectangular sections} \right]$$

$$\therefore \quad v_s(x) = \int_{x=0}^{x} \frac{3P}{2Gbh}\,dx + v_s(0) = \frac{3Px}{2Gbh} + 0$$

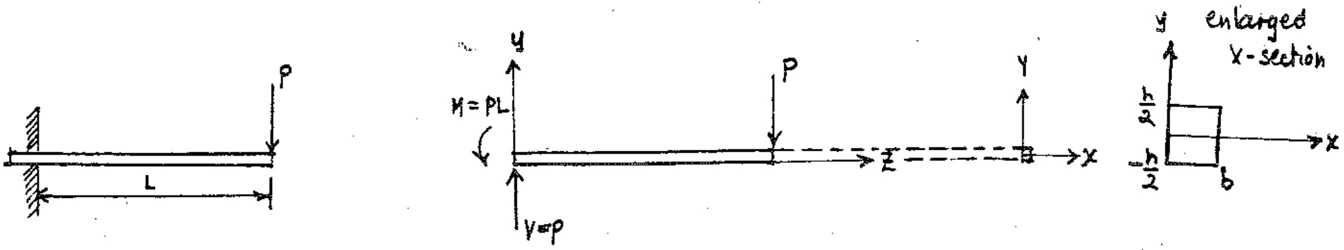
$$v_b(x) = \frac{Px^3}{3EI} + \frac{P(L-x)x^2}{2EI}$$, by method of superposition.

$$\frac{v_s(L)}{v_b(L)} = \frac{3PL/2Gbh}{PL^3/3EI} = \frac{9EI}{2GbhL^2} \quad ; \quad I = \frac{1}{12}bh^3; \quad \frac{E}{G} = 2(1+\nu)$$

$$\therefore \quad \left(\frac{v_s}{v_b}\right)_{x=L} = \frac{3(1+\nu)}{4}\frac{h^2}{L^2} \quad ; \quad \nu \leq 0.5 \text{ (From Thermodynamics)}$$

If $\frac{h}{L} \leq 10^{-1}$, $\left(\frac{v_s}{v_b}\right)_{x=L} \leq \frac{3 \times 1.5}{4} \times 10^{-2} = 1.025 \times 10^{-2}$, approx 1 percent.

(8.42)



$M_b = P[L-x]$

$V = P$ constant

From Ex. 8.10 $\quad v_b = \frac{PL^3}{3EI_{yy}}$ deflection due to bending

From eq. 7.32, energy due to shear $\quad U_s = \iiint \frac{\tau_{xy}^2}{2G}\,dx\,dy\,dz$

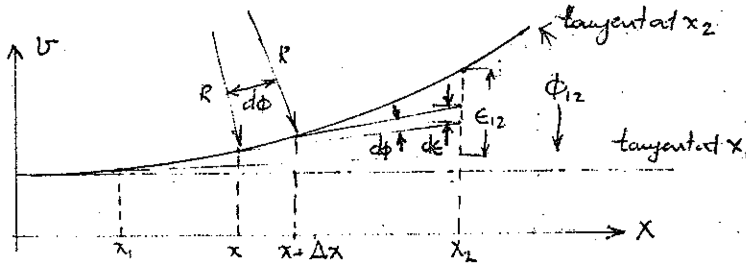
For rectangular beams $\quad \tau_{xy} = \frac{V}{2I_{yy}}\left[\left(\frac{h}{2}\right)^2 - y^2\right]$ (eq. 7.27)

$$U_s = \frac{P^2}{8GI_{yy}^2}\int_0^L \int_{-h/2}^{h/2} \int_0^b \left[\left(\frac{h}{2}\right)^4 - 2y^2\left(\frac{h}{2}\right)^2 + y^4\right] dx\,dy\,dz$$

$$\therefore, \quad U_s = \frac{b}{16}\frac{LP^2}{GI_{yy}^2}\frac{h^5}{15} = \frac{3LP^2}{4GI_{yy}}\frac{h^2}{15} \quad \text{where } I_{yy} = \frac{bh^3}{12} \quad ; \quad v_s = \frac{\partial U_s}{\partial P} = \frac{3}{2}\frac{LP}{GI_{yy}}\frac{h^2}{15}$$

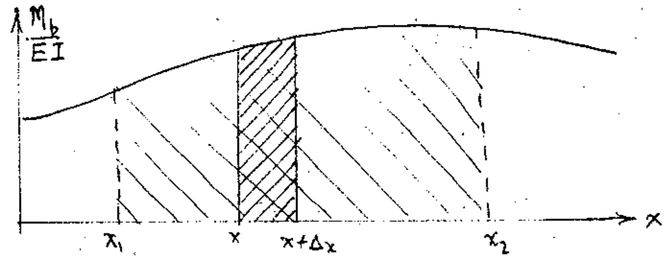
$\longrightarrow$ Finally $\left(\frac{v_s}{v_b}\right)_{x=L} = \frac{3}{5}(1+\nu)\frac{h^2}{L^2}$ where $G = \frac{E}{2(1+\nu)}$

8.43



(i) If the curvature is large, Δx may be taken as measured along the neutral axis. Suppose the radius of curvature is constant over dx. Then the angle between the tangents at the ends of dx is

$$d\phi = \frac{\Delta x}{R}$$



$$\frac{1}{R} \approx \frac{d^2v}{dx^2} = \frac{M_b}{EI}$$

$\therefore \; d\phi = \dfrac{M_b \Delta x}{EI}$ = shaded area in $\dfrac{M_b}{EI}$ diagram

Integrating, $\phi_{12} = \displaystyle\int_{x_1}^{x_2} \frac{M_b}{EI}dx$ = area of $\dfrac{M_b}{EI}$ diagram between x_1 and x_2.

(ii) From geometry, the increase $d\epsilon$ in ϵ due to an increment Δx is

$$d\epsilon = (x_2 - x)\,d\phi = (x_2 - x)\frac{M_b}{EI}\Delta x$$

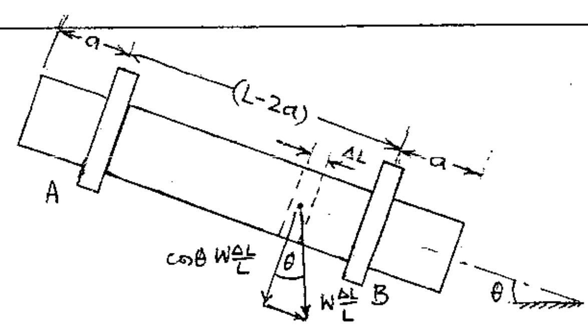
= first moment of shaded area about x_2.

$\epsilon_{12} = \displaystyle\int_{x_1}^{x_2} d\epsilon = \int_{x_1}^{x_2} (x_2 - x)\frac{M_b}{EI}dx$ = first moment of $\dfrac{M_b}{EI}$ diagram between x_1 and x_2, about x_2.

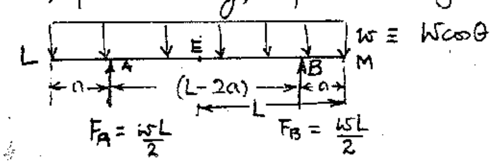
8.44 The loading at any point consists of:

(i) axial load $\frac{W}{L}\sin\theta$

(ii) normal load $\frac{W}{L}\cos\theta$

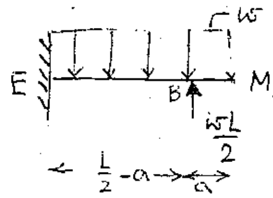


Thus, for bending, the free-body is



8.44 continued

Consider EM to be a cantilever fixed at E at zero slope. For the sections at L and M to remain parallel, they must remain parallel to the section at E, because the deflections of EL and EM are symmetrical about E.



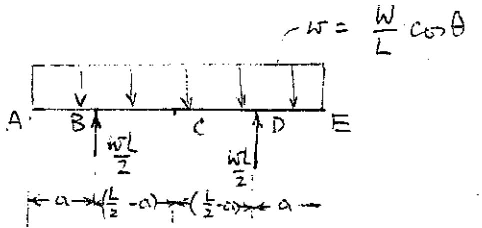
Slope at M (by superposition) is

$$\phi_M = -\frac{w\left(\frac{L}{2}\right)^3}{6EI} + \frac{wL}{2}\frac{\left(\frac{L}{2}-a\right)^2}{2EI} = 0$$

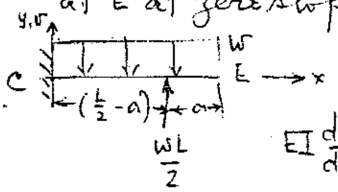
This gives $\underline{a = L\left(\frac{\sqrt{3}-1}{2\sqrt{3}}\right) = 0.211L}$

8.45 Arguing precisely as in 8.44, the free-body diagram, including only those forces that cause bending is:

Slope at C is zero from symmetry.



Consider CE as a cantilever fixed at E at zero slope:



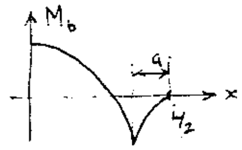
$$EI\frac{d^2v}{dx^2} = M_b = \frac{wL}{2}\langle \tfrac{L}{2}-a-x\rangle^1 - \frac{w}{2}\left(\tfrac{L}{2}-x\right)^2$$

Boundary Conditions: at $x=0$, $v = \frac{dv}{dx} = 0$.

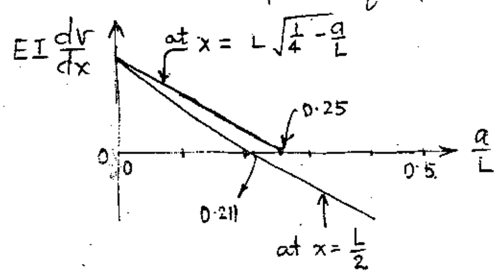
This gives

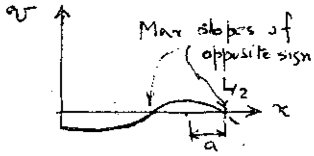
$$\frac{dv}{dx} = \frac{1}{EI}\left\{-\frac{wL}{4}\langle \tfrac{L}{2}-a-x\rangle^2 + \frac{w}{6}\left(\tfrac{L}{2}-x\right)^3 + \frac{3(L-2a)^2-L^2}{48}wL\right\}$$

$\left(\frac{dv}{dx}\right)_{max}$ occurs where $M_b = 0$ or $\begin{cases} x = L\sqrt{\frac{1}{4}-\frac{a}{L}}, & \frac{a}{L} < \frac{1}{4}, \text{ or } x = \frac{L}{2} \\ x = \frac{L}{2} \text{ for } \frac{a}{L} > \frac{1}{4} \end{cases}$



Consider a plot of $\frac{dv}{dx}$ at these two points against $\frac{a}{L}$.





8.45 continued

Suppose we take $\frac{a}{L}$ such that $\frac{dv}{dx}$ is of equal magnitude but of opposite sign at these two extrema.

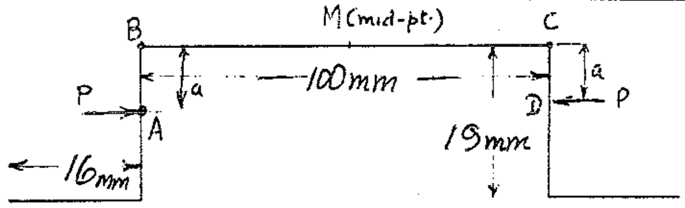
Then $0.211 < \frac{a}{L} < 0.25$

We see that if $\frac{a}{L}$ changes from this value, the max value of $\left|\frac{dv}{dx}\right|$ increases. Thus the above choice of $\frac{a}{L}$ gives $\min.\left[\left|\frac{dv}{dx}\right|_{max}\right]$.

By trial-and-error,

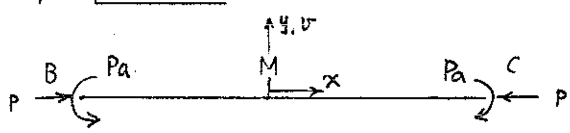
$$\underline{\frac{a}{L} \approx 0.22} \quad \text{in order that} \quad \left(\frac{dv}{dx}\right)_{x=L\sqrt{\frac{1}{4}-\frac{a}{L}}} = -\left(\frac{dv}{dx}\right)_{x=\frac{L}{2}}$$

8.46



$a = 10\,mm$

Consider a free-body of BC.



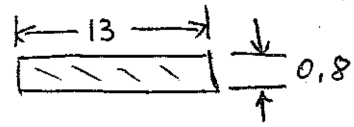
Assume bending effect of axial force is negligible ($\approx Pv$). Then

$$M_b = EI\frac{d^2v}{dx^2} = -Pa =$$

Boundary Conditions: $v = \frac{dv}{dx} = 0$ at $x = 0$

These give $v = -\frac{P}{EI}\frac{a}{2}x^2$, a parabola with vertex at M.

$$E = 205\times10^9\,N/m^2;\quad I = \frac{bh^3}{12} = \frac{13\times(0.8)^3}{12} = 0.554\times10^{-12}\,m^4$$



$$v = 4.40\times10^{-2}\,Px^2$$

at $x = 0.050$ $\qquad v_{max} = 1.1\times10^{-4}\,P$

if $v_{max} = 1\,mm = 10^{-3}\,m$

$P = 9\,N \quad (\approx 2\,LB)$

STRESS $(\sigma_x)_{max} = \frac{My}{I_{zz}} = 656\,MN/m^2$

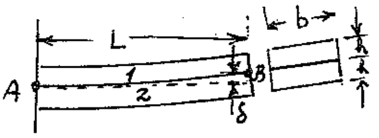
STRESS too HIGH; 1020 CR STEEL MIGHT BE USED

8.47

Reason as follows:

(i) As in ordinary beam theory, plane sections remain plane.

(ii) From a free-body, see that no net force or moment is transmitted across any section.

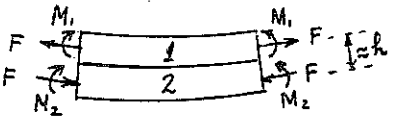


A, B are areas of stress concentration.

Conclude:

Stress and strain pictures do not vary along the beam. Forces transmitted by each beam individually are of equal magnitude, opposite sign.

Free-body:



For equilibrium, $M_1 + M_2 = Fh$, assuming uniform tensile stress across section.

Geometric Compatibility: strain and radius of curvature of common fibre at interface must be equal for the two beams.

Strains:

$$\epsilon_1 = \frac{F}{E_1 bh} + \frac{M_1 \frac{h}{2}}{E_1 I} + \alpha_1 T \qquad \text{------ (1)}$$

$$\epsilon_2 = \frac{F}{E_2 bh} + \frac{M_2 \frac{h}{2}}{E_2 I} + \alpha_2 T \qquad \text{------ (2)}$$

Curvatures:

$$\frac{1}{R_1} \approx \frac{M_1}{E_1 I} \qquad \frac{1}{R_2} \approx \frac{M_2}{E_2 I}\,, \qquad \text{where } I = \frac{bh^3}{12}$$

Since $R_1 = R_2$, $\quad M_1 = \frac{E_1}{E_2} M_2$. Also, $M_1 + M_2 = M_2\left(1 + \frac{E_1}{E_2}\right) = Fh$.

$$\therefore\ M_2 = \frac{Fh}{1 + \frac{E_1}{E_2}}, \quad M_1 = \frac{Fh}{1 + \frac{E_2}{E_1}}$$

Substituting in (1) and (2), and writing $\epsilon_1 = \epsilon_2$, we obtain F.

After some algebra,

$$M_2 = \frac{(\alpha_2 - \alpha_1)T}{\frac{1 + \frac{E_1}{E_2}}{E_1 bh^2} + \frac{1 + \frac{E_1}{E_2}}{E_2 bh^2} + \frac{6}{bh^2}\left(\frac{1}{E_1} + \frac{1}{E_2}\right)}$$

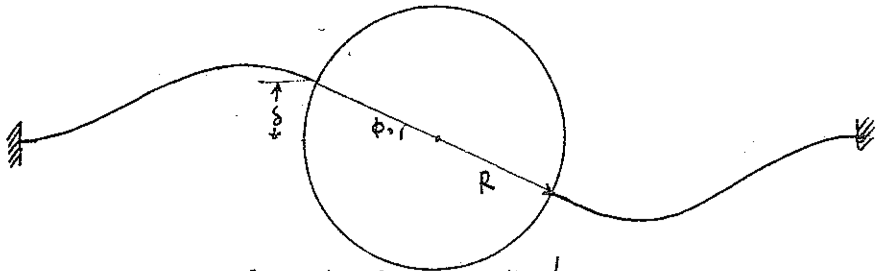
From geometry of circle,

$$\delta = \frac{L^2}{2R} = \frac{M_2 L^2}{2 E_2 I} = \frac{6 M_2 L^2}{E_2 bh^3}$$

This leads to

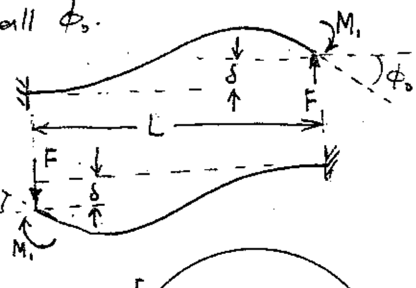
$$\delta = \frac{(\alpha_2 - \alpha_1) T L^2}{h} \cdot \frac{6}{14 + \frac{E_1}{E_2} + \frac{E_2}{E_1}}$$

8.48



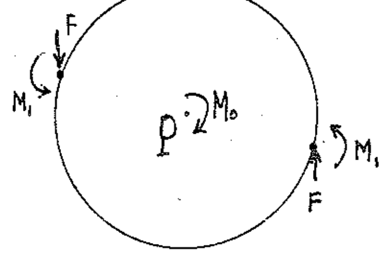
Geometric Compatibility: $\delta \approx R\phi_0$, for small ϕ_0.

Forces on strips and disc:



Equilibrium of disc

$$\Sigma M_p = 0: \quad M_0 = 2FR + 2M_1 \quad ----(1)$$



Deformation of strips, method of superposition:

$$\delta = \frac{FL^3}{3EI} - \frac{M_1 L^2}{2EI}$$

$$\phi_0 = \frac{M_1 L}{EI} - \frac{FL^2}{2EI} \quad \text{and} \quad \delta = R\phi_0$$

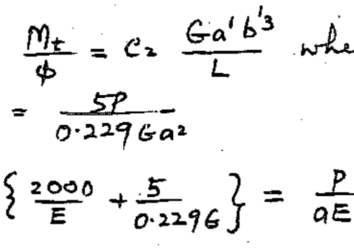
Using these three equations and (1), obtain

$$F = \frac{M_0}{2R + \frac{2}{3}L \times \left(\frac{2L+3R}{L+2R}\right)}, \quad \phi_0 = -\frac{FL^2}{2EI} + \frac{FL^2}{3EI} \cdot \frac{2L+3R}{L+2R}$$

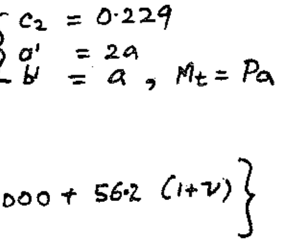
$$K \equiv \frac{M_0}{\phi_0} = \frac{EI}{L^2} \cdot \frac{2R + \frac{2}{3}L \times \left(\frac{2L+3R}{L+2R}\right)}{-\frac{1}{2} + \frac{1}{3} \cdot \frac{2L+3R}{L+2R}} = \frac{8EI}{L^3}\left(3R^2 + 3RL + L^2\right)$$

8.49

Bending Deflection at A is same as at B.



is equivalent to:



$$(\delta_A)_{bending} = \frac{P(10a)^3}{3E\frac{1}{2}\cdot 2a \cdot a^3} = \frac{2000P}{aE}$$

Torsion of shaft through angle ϕ causes A to deflect through

$$(\delta)_{torsion} = a\phi.$$

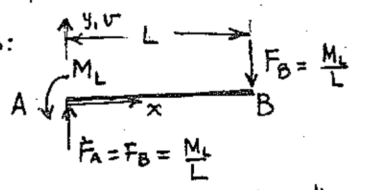
Eqn. 6.26 (p 265) is equivalent to $\frac{M_t}{\phi} = c_2 \frac{Ga'b'^3}{L}$ where $\begin{cases} c_2 = 0.229 \\ a' = 2a \\ b' = a, \quad M_t = Pa \end{cases}$

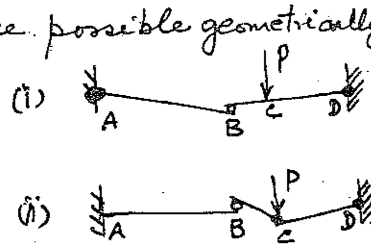
This gives $\phi = \frac{Pa \cdot 10a}{0.229G \cdot 2a \cdot a^3} = \frac{5P}{0.229Ga^2}$

$$\therefore \delta_A = (\delta)_{bending} + (\delta)_{torsion} = \frac{P}{a}\left\{\frac{2000}{E} + \frac{5}{0.229G}\right\} = \frac{P}{aE}\left\{2000 + 56.2\,(1+\nu)\right\}$$

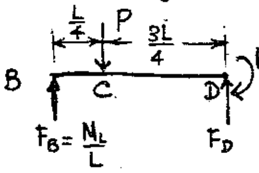
8.50 Two collapse mechanisms are possible geometrically:

Free-Bodies:
Case (i)





Note that at A, $M_b = M_L$ in order that a plastic hinge should exist there.



$\sum M_{Dz} = 0: \quad M_D = P \cdot \frac{3L}{4} - F_B \cdot L = \frac{3PL}{4} - M_L$

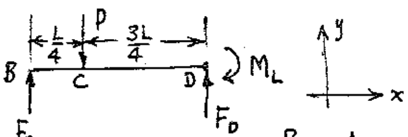
But we must have $M_D = M_L$ as a plastic hinge exists at D. $\therefore \quad M_L = \frac{3PL}{4} - M_L$ or $\underline{\underline{P = \frac{8M_L}{3L}}}$

Check and see whether $M_c < M_L$ (Otherwise a plastic hinge will occur there before both A and D can be plastic simultaneously.)

$M_c = F_B \frac{L}{4} = \frac{M_L}{4} < M_L$. The configuration in Fig (i) is possible

(continued.....)

Case (ii) Free-body of BD:



Bending Moment at $D = M_L$ because D is a plastic hinge

Equilibrium: $\sum F_y = 0$:

$F_B + F_D = P$ ---------- (i)

$\sum M_{Dz} = 0: \quad M_L = \frac{3PL}{4} - F_B L$ ----- (ii)

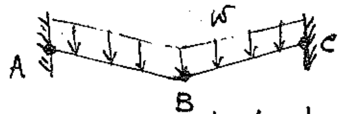
For a plastic hinge at C, M_c must be equal to M_L.

$M_c = F_B \frac{L}{4} = M_L \quad \therefore \quad F_B = \frac{4M_L}{L}$ and from (ii) $P = \frac{20}{3}\frac{M_L}{L} > \frac{8M_L}{3L}$

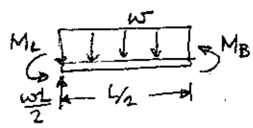
Therefore, collapse will occur as in case (i), because occurs at the lowest possible load. Also, a check shows that $M_A = 4M_L$ for case (ii), which means that the configuration in Fig (ii) is not possible.

$$P_L = \frac{8M_L}{3L}$$

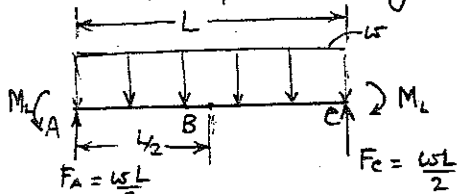
8.51 Any collapse mechanism must be symmetrical about the centre of the beam. The only geometrically compatible mechanism is:



In the free-body sketched, the moments at A, C are M_L because A & C are plastic hinges.



A, B, C: plastic hinges



FREE BODY

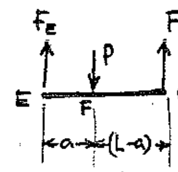
Bending Moment at B:

$$M_B = \frac{wL}{2}\cdot\frac{L}{2} - M_L - \left(\frac{wL}{2}\right)\frac{L}{4}$$

$$= \frac{wL^2}{8} - M_L$$

The requirement for B to be a plastic hinge is $|M_b| = M_L$. This gives $\frac{wL^2}{8} - M_L = M_L$, or $\underline{\underline{w = \frac{16 M_L}{L^2}}}$

8.52 Assume swing is rigid. Then the free-body of the swing is:



The forces F_E, F_G are transmitted to the beam AD unchanged. $\therefore$

$$F_B = F_E = P\frac{a}{L}$$

$$F_C = F_G = P\left(\frac{L-a}{L}\right).$$

$$F_E = P\frac{a}{L}$$

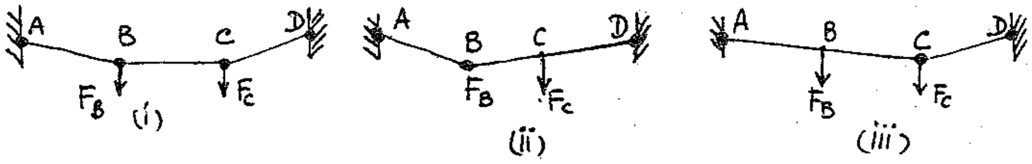
$$F_G = P\left(\frac{L-a}{L}\right)$$

We may reason that the collapse mechanism actually occurring will have the minimum

CONTINUED

8.52 CON'T

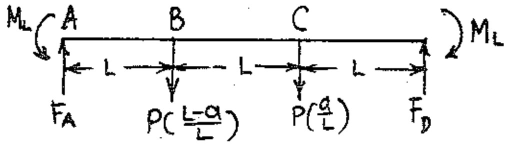
number of plastic hinges required for geometric compatibility.
Possible Collapse Mechanisms:



Our reasoning tells us that either (ii) or (iii) will occur (i), unless $M_B = M_C$, the symmetric case, when $a = \frac{L}{2}$.

When $a \neq \frac{L}{2}$: Free-body of AD

From Equilibrium requirements,



$$F_A = \frac{P(2L-a)}{3L}, \quad F_D = \frac{P(L+a)}{3L}$$

Collapse Mechanism (ii): require $M_B = M_L$.

ie, $M_B = F_A L - M_L = \frac{P(2L-a)}{3} - M_L = M_L$. $\therefore$ $$P = \frac{6M_L}{2L-a}$$

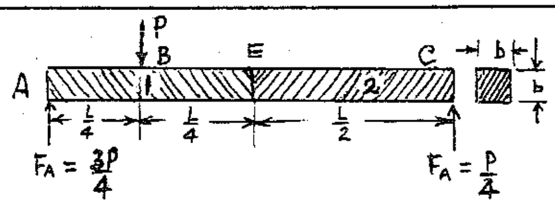
Collapse Mechanism (iii) require $M_C = M_L$

ie, $M_C = F_D L - M_L = \frac{P(L+a)}{3} - M_L = M_L$ $\therefore$ $$P = \frac{6M_L}{L+a}$$

$\therefore$ If $a \geq \frac{L}{2}$, $P_L = \frac{6LM_L}{L+a}$

If $a \leq \frac{L}{2}$, $P_L = \frac{6LM_L}{2L-a}$

8.53



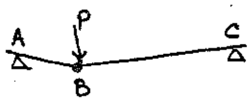
$K = \frac{M_L}{M_Y} = 1.5$

By definition,

$$M_{Y1} = \frac{b^3 Y_1}{6} = \frac{b^3 Y}{6}$$

$$M_{Y2} = \frac{b^3 Y_2}{6} = \frac{b^3 Y}{12}$$

$M_{L1} = \frac{b^3 Y}{4}$; $M_{L2} = \frac{b^3 Y}{8}$. A possible collapse Mechanism:



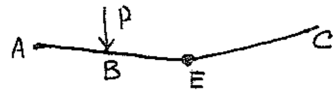
Require $M_B = M_L$. This gives $F_A \cdot \frac{L}{4} = \frac{3PL}{16} = M_{L1}$, or $P = \frac{16 M_{L1}}{3L}$

or $P = \frac{4}{3}\frac{b^3 Y}{L}$. Bending Moment at $x = \frac{L}{2} = \frac{4}{3}\frac{b^3 Y}{L} \cdot \frac{L}{2} = \frac{2}{3} b^3 Y > \frac{b^3 Y}{8}$

CONTINUED

8.53 continued

Thus a plastic hinge will occur in material ②, right next to the centre of the beam:



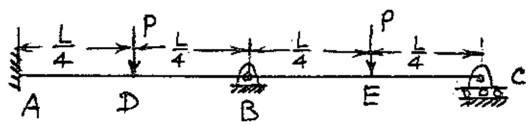
$\therefore\ M_E = M_{L2}$.

$$M_E = F_A \cdot \frac{L}{2} = \frac{3P}{4} \cdot \frac{L}{2} = \frac{3PL}{8} = M_{L2} = \frac{b^3 Y}{8}$$

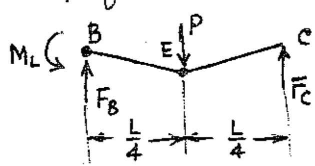
$$\therefore\ \underline{\underline{P_L = \frac{b^3 Y}{3L}}}$$

8.54

Two Mechanisms:



(i) Suppose plastic hinges occur at B and E, Draw a free-body of BC:

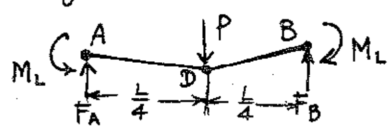


$\sum M_{Bz} = 0$: gives $F_C = \frac{P}{2} - \frac{M_L}{L}$

Require $M_E = M_L$.

$$M_E = F_C \cdot \frac{L}{4} = \frac{PL}{8} - \frac{M_L}{4} = M_L \quad \therefore\ \underline{\underline{P = \frac{10 M_L}{L}}}$$

(ii) Suppose plastic hinges occur at A, D and B: draw a free-body of AB:



Equilibrium requires

$$F_A = \frac{P}{2} = F_B.$$

Require $M_D = M_L$.

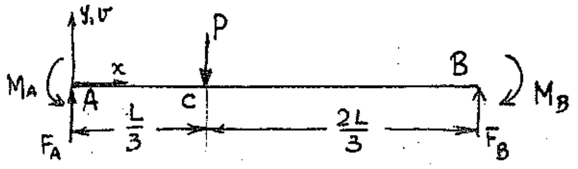
ie., $M_D = F_A \cdot \frac{L}{4} - M_L = \frac{PL}{8} - M_L = M_L$

This gives $P = \frac{16 M_L}{L} > \frac{10 M_L}{L}$

Thus collapse mechanism (i) actually occurs, with plastic hinges at B and E,

$$P_L = \frac{10 M_L}{L}.$$

8·55 Elastic Analysis to find P_Y:



FREE BODY

Equilibrium: $\sum F_y = 0$:

$$F_A + F_B = P \quad \text{------} \quad (1)$$

$\sum M_{Bz} = 0$: $F_A \cdot L + M_B = M_A + \frac{2PL}{3}$ ---- (2)

Bending Moment: $EI\frac{d^2v}{dx^2} = M_b = F_A x - P\langle x - \frac{L}{3}\rangle^1 - M_A$

$$\therefore EI\frac{dv}{dx} = \frac{F_A}{2}x^2 - \frac{P}{2}\langle x - \frac{L}{3}\rangle^2 - M_A x + C_1$$

$$EIv = \frac{F_A}{6}x^3 - \frac{P}{6}\langle x - \frac{L}{3}\rangle^3 - \frac{M_A}{2}x^2 + C_1 x + C_2$$

Boundary Conditions: at $x = 0$, $v = \frac{dv}{dx} = 0$

$x = L$ $\quad v = \frac{dv}{dx} = 0$

These, along with (1) and (2) give $C_1 = C_2 = 0$,

$$F_A = \frac{20}{27}P \qquad M_A = \frac{4}{27}PL$$

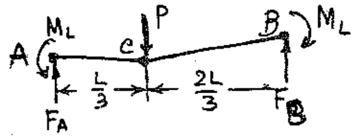
$$F_B = \frac{7}{27}P \qquad M_B = \frac{2}{27}PL$$

$$M_c = -\frac{4}{27}PL + \frac{20}{27}P\frac{L}{3} = \frac{8PL}{81}$$

$$\text{Max}(M_A, M_B, M_c) = M_A = \frac{4}{27}PL$$

$\therefore$ Yielding occurs first at A. $M_A = M_Y = \frac{4}{27}P_Y L \quad \therefore \underline{P_Y = \frac{27}{4}\frac{M_Y}{L}}$

Collapsed Free-Body:



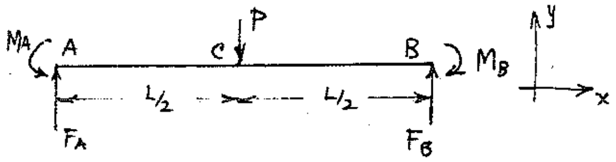
Equilibrium: $F_A = \frac{2P}{3}$,

$F_B = \frac{P}{3}$. $\quad$ Require $M_c = M_L$

i.e., $M_c = F_A \cdot \frac{L}{3} - M_L = \frac{2PL}{9} - M_L = M_L$

$$\therefore \underline{P_L = \frac{9M_L}{L}} \quad \text{and} \quad \frac{P_L}{P_Y} = \frac{9M_L}{\frac{27}{4}M_Y} = \frac{4}{3}K$$

8-56 Elastic Analysis to find P_Y:



Equilibrium:

$\sum F_y = 0$: $\quad F_A + F_B = P$ ----- (1)

$\sum M_{Bz} = 0$: $\quad F_A \cdot L + M_A = P\frac{L}{2} + M_B$ --- (2)

Bending Moment: $EI\frac{d^2v}{dx^2} = M_b = F_A x - P\langle x - \frac{L}{2}\rangle^1 - M_A$

$$\therefore EIv = \frac{F_A}{6}x^3 - \frac{P}{6}\langle x - \frac{L}{2}\rangle^3 - \frac{M_A}{2}x^2 + C_1 x + C_2$$

Boundary Conditions: at $x = 0$ and $x = L$, $v = \frac{dv}{dx} = 0$

CONTINUED

8.56

These give, along with (1) and (2), $F_A = F_B = P/2$

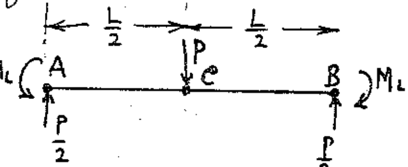
$M_A = M_B = \frac{PL}{8}$

Bending Moment at $C = \frac{P}{2}\cdot\frac{L}{2} - M_A = \frac{PL}{8}$.

$\therefore$ $M_A = M_B = M_C$. For yielding, $\frac{PL}{8} = M_y$ $\therefore$ $\underline{\underline{P_y = \frac{8M_y}{L}}}$

<u>Collapse</u>: Plastic hinges occur at A, B and C at any load P, $M_A = M_B = M_C$. Therefore M_L is reached simultaneously at A, B, C.

Free-body:

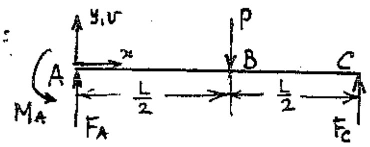


Require $M_C = M_L$, ie, $M_C = \frac{P}{2}\cdot\frac{L}{2} - M_L = M_L$ or $\underline{\underline{P_L = \frac{8M_L}{L}}}$

$\therefore$ $\frac{P_L}{P_y} = \frac{8M_L}{8M_y} = K$

8.57

Free-Body:



Elastic Analysis for P_y:

Equilibrium:

$\Sigma F_y = 0$: $F_A + F_C = P$ ------ (1)

$\Sigma M_{cz} = 0$: $M_A + F_A \cdot L = P\frac{L}{2}$ ------ (2)

Bending Moment:

$EI\frac{d^2v}{dx^2} = M_b = F_A x - M_A - P\langle x - \frac{L}{2}\rangle^1$.

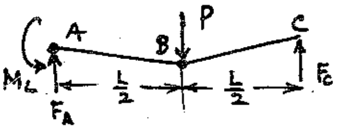
Boundary Conditions: at $x=0$, $v = \frac{dv}{dx} = 0$; at $x = L$, $v = 0$.

Obtain: $F_A = \frac{5P}{8}$, $F_C = \frac{3P}{8}$, $M_A = \frac{PL}{8}$

Bending Moment is max. at B : $M_b = \frac{3PL}{16}$. Yielding occurs here first.

$\therefore$ $\frac{3PL}{16} = M_y$ or $\underline{P_y = \frac{16M_y}{3L}}$

Collapse occurs with plastic hinges at A, B.



From equilibrium, $F_A = \frac{P}{2} + \frac{M_L}{L}$; $F_C = \frac{P}{2} - \frac{M_L}{L}$.

Require $M_B = M_L$, ie, $F_C\frac{L}{2} = \frac{PL}{4} - \frac{M_L}{2} = M_L$, or $\underline{\underline{P_L = \frac{6M_L}{L}}}$ $\therefore$ $\frac{P_L}{P_y} = \frac{9}{8}\frac{M_y}{M_L} = \frac{9}{8}K$

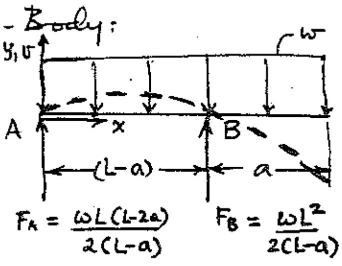
8.58 Reason as follows:

(i) No part of the rod is in continuous contact with the table, because $L < (1+\sqrt{2})a$

(ii) for equilibrium, the rod must touch the table at two points, one of them at the edge.

(iii) the second point must be an end of the rod.

Free-Body:



Bending Moment:

$$EI\frac{d^2v}{dx^2} = M_b = \frac{wL(L-2a)}{2(L-a)}x + \frac{wL^2}{2(L-a)}\langle x+a-L\rangle^1 - \frac{wx^2}{2}$$

$$\therefore EIv = \frac{wL(L-2a)}{2(L-a)}\frac{x^3}{6} + \frac{wL^2}{2(L-a)}\frac{\langle x+a-L\rangle^3}{6} - \frac{wx^4}{24} + C_1x + C_2$$

Boundary Conditions: $v = 0$ at $x = 0$, $x = L-a$.

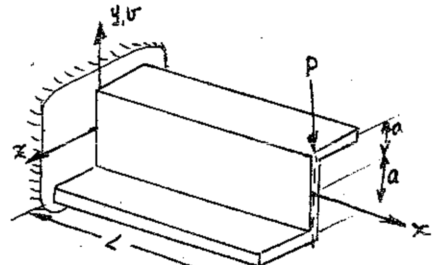
These give

$$v = \frac{1}{EI}\left\{\frac{wL(L-2a)x^3}{12(L-a)} + \frac{wL^2}{12(L-a)}\langle x-(L-a)\rangle^3 - \frac{wx^4}{24} - \frac{wx(L^3-3aL^2+a^2L+a^3)}{24}\right\}$$

8.59 Bending Moments – Curvature

$$\frac{d^2v}{dx^2} = \frac{1}{E}\,\frac{I_{yz}M_{by} + I_{yy}M_{bz}}{I_{yy}I_{zz} - I_{yz}^2}$$

$$\frac{d^2w}{dx^2} = -\frac{1}{E}\,\frac{I_{zz}M_{by} + I_{yz}M_{bz}}{I_{yy}I_{zz} - I_{yz}^2}$$



In this case, $M_{by} = 0$, $M_{bz} = -P(L-x)$

$I_{yy} = \frac{8}{3}a^3t$; $I_{yz} = -a^3t$; $I_{zz} = \frac{2}{3}a^3t$

$$\therefore \frac{d^2v}{dx^2} = -k_1P(L-x) \quad \text{where } k_1 \equiv \frac{I_{zz}}{E(I_{yy}I_{zz}-I_{yz}^2)}$$

$$\frac{d^2w}{dx^2} = k_2P(L-x) \quad \text{where } k_2 \equiv \frac{I_{yz}}{E(I_{yy}I_{zz}-I_{yz}^2)}$$

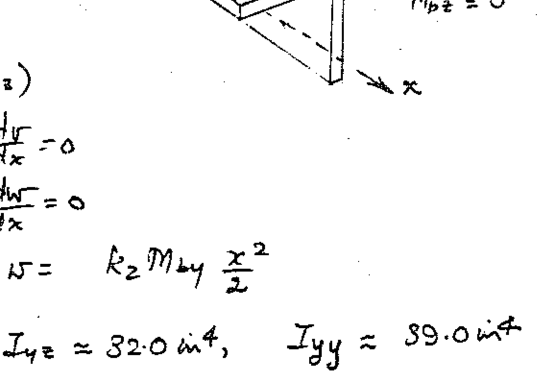
Boundary Conditions: at $x=0$: $v = \frac{dv}{dx} = 0$, $w = \frac{dw}{dx} = 0$

These give $v = \frac{k_1P}{6}\left[-(L-x)^3 - 6x\frac{L^2}{2} + L^3\right]$ at $x = L$ $v = -\frac{k_1PL^3}{3} = -\frac{2PL^3}{7Eta^3}$

$w = \frac{k_2P}{6}\left[(L-x)^3 + \frac{6xL^2}{2} - L^3\right]$ $\qquad w = \frac{k_2PL^3}{3} = \frac{3PL^3}{7Eta^3}$

8.60

The equations of Prob 8.59 become



$$\frac{d^2v}{dx^2} = k_1M_{by} \quad \text{where } k_1 \equiv \frac{I_{yz}}{E(I_{yy}I_{zz}-I_{yz}^2)}$$

$$\frac{d^2w}{dx^2} = k_2M_{by} \quad \text{where } k_2 \equiv \frac{I_{zz}}{E(I_{yy}I_{zz}-I_{yz}^2)}$$

Boundary Conditions: at $x=0$, $v = \frac{dv}{dx} = 0$

$w = \frac{dw}{dx} = 0$

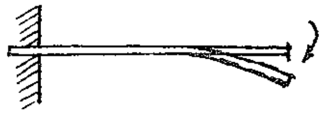
These give $v = k_1M_{by}\frac{x^2}{2}$, $\quad w = k_2M_{by}\frac{x^2}{2}$

For this problem, $I_{zz} \approx 60\text{ in}^4$, $I_{yz} \approx 32.0\text{ in}^4$, $I_{yy} \approx 89.0\text{ in}^4$

at $x = 10' = 120''$, get

$v = 1.43''$ $\qquad w = -2.62''$

8.61)



This is the case of non-linear deflection into circle of radius R.

From section 8.2 $\quad \frac{1}{R} = \frac{M_0}{EI} = \frac{v''}{[1+v'^2]^{3/2}} \quad v' = \frac{dv}{dx}$

Let $p = \frac{dv}{dx} \quad p' = \frac{d^2v}{dx^2} = \frac{dp}{dx} \quad K = \frac{1}{R} = \frac{p'}{[1+p^2]^{3/2}}$

non-dimensionalize R + x w.r.t. L

L = Length of beam

$R' = \frac{R}{L} \; ; \; x' = \frac{x}{L}$

$\frac{dp}{dx} = \frac{dp}{L\,dx'}$

$$K^* = \frac{1}{R'} = \frac{L}{R} = \frac{\frac{dp}{dx'}}{[1+p^2]^{3/2}}$$

$$K^* x' = \int_0^p \frac{dp}{[1+p^2]^{3/2}} = \frac{p}{\sqrt{1+p^2}} \qquad (1)$$

Rewriting (1), we get

$$p^2\left[1-(K^*x')^2\right] = (K^*x')^2 \qquad (2)$$

If $K^* \ll 1$ i.e. $R \gg L$

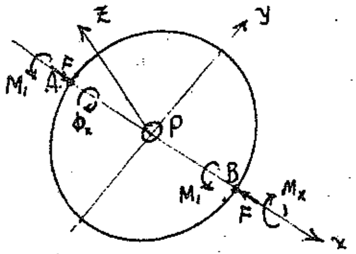
$$p = K^* x' = LK\frac{x}{L} = Kx$$

$$\therefore \frac{dv}{dx} = \frac{M_0}{EI}x \quad \text{or} \quad v = \frac{M_0 x^2}{EI}$$

Note this result agrees with case 3 in Table 8.1

8·62 (a)

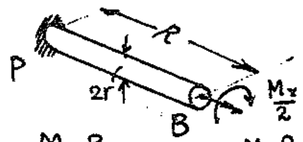
Free-body of Rim



Equilibrium:

$$M_1 = \frac{M_x}{2}.$$

Torsion of cross-arm PB:

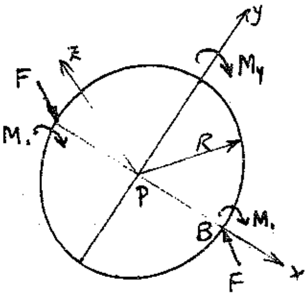


$$\phi_x \equiv \phi_B = \frac{M_x R}{G\frac{\pi}{2}r^4} = \frac{M_x R}{\pi G r^4} \qquad \therefore \; K_x \equiv \frac{M_x}{\phi_x} = \frac{\pi G r^4}{R} = \frac{\pi E r^4}{2R(1+\nu)},$$

where $G = \dfrac{E}{2(1+\nu)}$

(b)

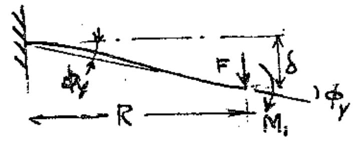
Free-body of Rim



Equilibrium:

$$M_y = 2RF - 2M_1 \quad \text{——(1)}$$

Deflection of cross-arm PB:



By superposition,

$$\delta = \frac{FR^3}{3EI} - \frac{MR^2}{2EI} \quad \text{——(2)}$$

$$\phi_y = \frac{FR^2}{2EI} - \frac{MR}{EI} \quad \text{——(3)}$$

Geometric Compatibility:

$$\delta = R\phi_y \quad \text{——(4)}$$

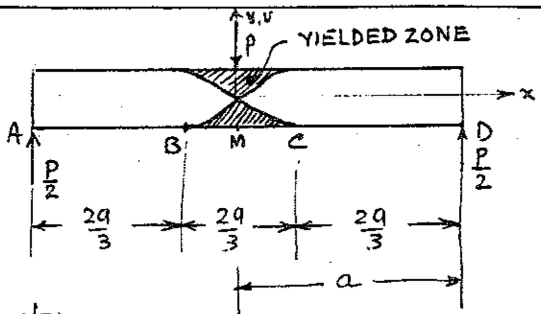
From (1), (2), (3), (4),

$$F = \frac{6EI\phi_y}{R^2}, \quad M_1 = \frac{2EI\phi_y}{R}, \quad M_y = \frac{8EI\phi_y}{R}$$

$$K_y \equiv \frac{M_y}{\phi_y} = \frac{8EI}{R}$$

$$\text{or} \quad = \frac{2\pi E r^4}{R} \qquad \left[\text{Note: } \frac{K_y}{K_x} = 4(1+\nu) \leq 6\right]$$

8·63 Moment-Curvature Relations:



$$M_b = \frac{3}{2}M_Y\left\{1 - \frac{1}{3}\left[\frac{(1/\rho)_Y}{(1/\rho)}\right]^2\right\}, \quad |x| \leq \frac{a}{3} \quad \text{----(1)}$$

$$M_b = \frac{EI_{YY}}{\rho}, \quad \frac{a}{3} < |x| \leq a \quad \text{---(2)}$$

ρ = Radius of Curvature

8-63 continued

We also have: $M_y = \frac{bh^2}{6} Y$ (7.51, p 328)

$$\frac{1}{\rho} \approx \frac{d^2v}{dx^2} \quad (8.3,\ p\ 362)$$

$$\left(\frac{1}{\rho}\right)_Y = \frac{\epsilon_Y}{h/2} = \frac{2Y}{Eh} \quad (7.52,\ p\ 329)$$

For the free-body,

$$M_b = \frac{P(a-x)}{2} \qquad 0 < x < a$$

Substituting in (1) and (2),

$$\frac{P(a-x)}{2} = \frac{3}{2} M_Y \left\{ 1 - \frac{1}{3} \left[\frac{2Y/Eh}{\frac{d^2v}{dx^2}} \right]^2 \right\} \quad , |x| \leq \frac{a}{3} \quad ---- \quad (3)$$

$$\frac{P(a-x)}{2} = EI \frac{d^2v}{dx^2} \qquad \frac{a}{3} \leq |x| \leq a \quad ---- \quad (4)$$

When the plastic regions have joined at the neutral axis, $(M_b)_{x=0} = 1.5 M_y = M_L$, from Table 7.1.

$$\therefore \frac{Pa}{2} = 1.5 M_y \quad \text{or} \quad P = \frac{3M_y}{a}$$

Then equations (3) and (4) may be written:

$$\sqrt{\tfrac{1}{3}} \left[\frac{2Y/Eh}{\frac{d^2v}{dx^2}} \right] = \left(\frac{x}{a}\right)^{\frac{1}{2}} \qquad |x| \leq \frac{a}{3} \quad ---- \quad (5)$$

and

$$EI \frac{d^2v}{dx^2} = \frac{3M_y}{2a}(a-x) \qquad \frac{a}{3} < |x| < a \quad ---- \quad (6)$$

The boundary conditions for (5) and (6) are:

at $x = 0$, $\frac{dv}{dx} = 0$

at $x = a$, $v = 0$

at $x = \pm\frac{a}{3}$, $(v, \frac{dv}{dx})$ from (5) $=$ $(v, \frac{dv}{dx})$ from (6)

These give (after much algebra)

$$v = -\frac{20}{27}\frac{a^2 M_y}{EI} + \frac{2\sqrt{a}\,Y}{\sqrt{3}\,Eh}\,\frac{4}{3}\,x^{3/2} \qquad |x| \leq \frac{a}{3}$$

$$v = \frac{1}{EI}\left\{ (a-x)\, M_y a \left[\frac{1}{4}\left(1-\frac{x}{a}\right)^2 - 1 \right] \right\} \qquad \frac{a}{3} \leq |x| \leq a$$

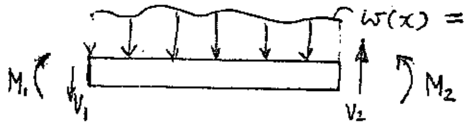
$Y = \frac{6M_y}{bh^2}$, $I = \frac{bh^3}{12}$ Using these, get the required form:

$$v = -\frac{a^2 M_y}{EI}\left[\frac{20}{27} - \frac{4}{3\sqrt{3}}\left(\frac{x}{a}\right)^{3/2} \right] \qquad |x| \leq \frac{a}{3}$$

$$v = -\frac{M_y a^2}{4EI}\left(3 - \frac{x}{a}\right)\left[1 - \left(\frac{x}{a}\right)^2 \right] \qquad \frac{a}{3} \leq |x| \leq a$$

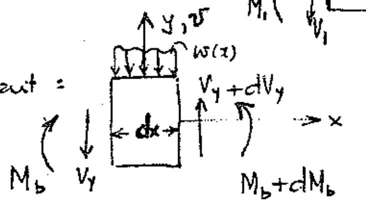
8.64

Free-body:



$w(x) = +k\,v(x) - q(x)$

Element :



Equilibrium:

$\Sigma F_y = 0: \quad dV_y = w(x)\,dx = +kv(x)\,dx - q(x)\,dx$

$\Sigma M_z = 0: \quad V_y\,dx = -dM_b$

In the limit, $\quad \dfrac{dM_b}{dx} = -V_y \quad , \quad \dfrac{d^2M_b}{dx^2} = -\dfrac{dV_y}{dx} = q(x) - kv(x)$

For Beams in bending, $\quad M_b = EI\dfrac{d^2v}{dx^2}$

$$\therefore \quad EI\frac{d^4v}{dx^4} + kv(x) = q(x) \quad ------(1)$$

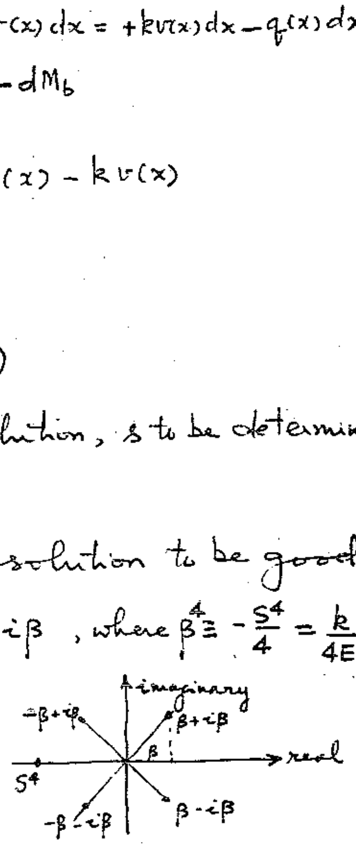
For the case $q(x) = 0$, try $\quad v = e^{sx}$ as a solution, s to be determined.

Substituting, it is necessary that

$(EIs^4 + k)\,v = 0 \quad$, for the solution to be good.

This gives $\quad s^4 = -\dfrac{k}{EI} \quad , \quad$ or $\quad s = \pm\beta \pm i\beta$, where $\beta^4 \equiv -\dfrac{s^4}{4} = \dfrac{k}{4EI}$

$\therefore$ The four independent solutions are



$v = e^{(\beta + i\beta)x}, \quad v = e^{(\beta - i\beta)x}, \quad v = e^{(-\beta + i\beta)x}, \quad v = e^{(-\beta - i\beta)x}$

Because the differential equation (1) is linear,
any linear combination of these is a solution;

$$v = Ae^{(\beta+i\beta)x} + Be^{(\beta-i\beta)x} + Ce^{(-\beta+i\beta)x} + De^{(-\beta-i\beta)x}, \quad A, B, C, D : \text{arbitrary constants}$$

$$= e^{\beta x}\left[(A+B)\cos\beta x + (A-C)\sin\beta x\right]$$

$$+ e^{-\beta x}\left[(C+D)\cos\beta x + (C-D)\sin\beta x\right]$$

This is equivalent to

$$v = e^{-\beta x}(C_1\cos\beta x + C_2\sin\beta x) + e^{\beta x}(C_3\cos\beta x + C_4\sin\beta x)$$

For a very long beam with a point load at $x = 0$,

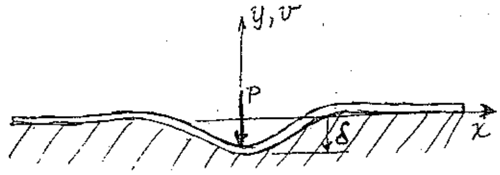
$$v, \ \frac{dv}{dx} \rightarrow 0 \ \text{ as } \ x \rightarrow \infty$$

CONTINUED

8.64 continued

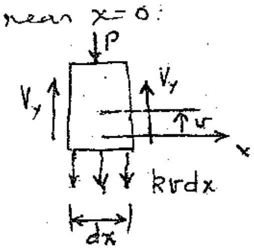
$\therefore$ For $x>0$, we require $C_3 = C_4 = 0$.

$\therefore\quad v = e^{-\beta x}(C_1 \cos\beta x + C_2 \sin\beta x)$



Boundary Conditions at $x=0$: $\dfrac{dv}{dx} = 0$

Another condition is obtained for the shear stress very near $x=0$:



$$2V_y + kv\,dx = P$$

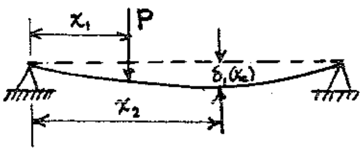
$$\text{as } dx \to 0,\quad V_y = \frac{P}{2}$$

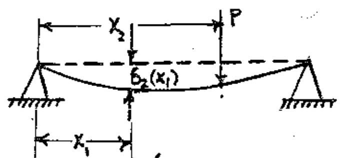
$$\text{But } V_y = -EI\frac{d^3 v}{dx^3}$$

Using these two boundary conditions, $C_1 = C_2 = -\dfrac{P\beta}{2k}$

and $\quad \delta \equiv -v(0) = -C_1 = +\dfrac{P\beta}{2k} = \dfrac{P}{2k}\left(\dfrac{k}{4EI}\right)^{\frac{1}{4}} = \dfrac{P}{(64EIk^3)^{1/4}}$

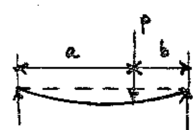
8.65)





From Table 8.1 case 4

$$\delta = \frac{Pb}{6LEI}\left[\frac{L}{b}\langle x-a\rangle^3 - x^3 + (L^2 - b^2)x\right]$$



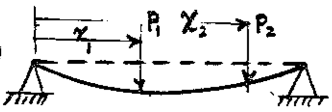
(a) (case 1) (case 2)

Case 1: $\delta_1(x_2) = \dfrac{P(L-x_1)}{6LEI}\left[\dfrac{L}{(L-x_1)}\langle x_2 - x_1\rangle^3 - x_2^3 + (L^2 - (L-x_1)^2)x_2\right] = \dfrac{P}{6LEI}\Big[-Lx_1\{3x_2^2 + x_1^2\} + 2L^2 x_1 x_2 + x_1 x_2(x_2^2 + x_1^2)\Big]$

Case 2: $\delta_2(x_1) = \dfrac{P(L-x_2)}{6LEI}\left[-x_1^3 + (L^2 - (L-x_2)^2)x_1\right] = \dfrac{P}{6LEI}\left[-Lx_1\{3x_2^2 + x_1^2\} + 2L^2 x_1 x_2 + x_1 x_2(x_2^2 + x_1^2)\right]$

Hence $\delta_1(x_2) = \delta_2(x_1)$

(b)



$$\left.\begin{aligned} M_b &= R_A x - P_1\langle x - x_1\rangle^1 - P_2\langle x - x_2\rangle^1 \\ R_A L &= P_1(L - x_1) + P_2(L - x_2) \end{aligned}\right\} M_b = P_1(1-x_1)x + P_2(1-x_2)x - P_1\langle x - x_1\rangle^1 - P_2\langle x - x_2\rangle^1$$

x is non-dimension. w.r.t. L

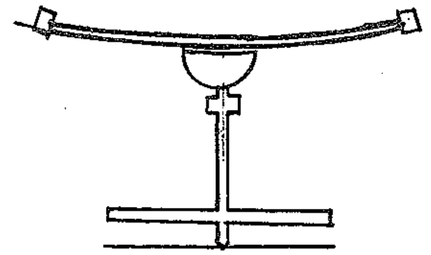
Let $M_b = P_1 f_1(x) + P_2 f_2(x)$; $U = \displaystyle\int_0^1 \frac{M_b^2 L^3}{2EI}dx = L^3\int_0^1 \frac{[P_1 f_1(x) + P_2 f_2(x)]^2}{2EI}dx$

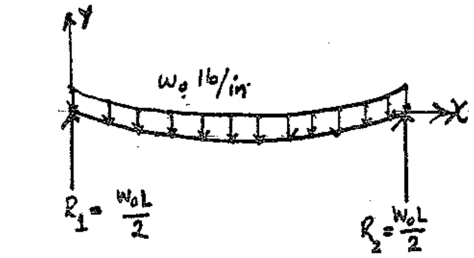
$$\delta_2(x_1) = \left.\frac{\partial U}{\partial P_2}\right|_{P_2=0} = \frac{P_1 L^3}{EI}\int_0^1 f_2 f_1\,dx$$

$$\delta_1(x_2) = \left.\frac{\partial U}{\partial P_1}\right|_{P_1=0} = \frac{P_2 L^3}{EI}\int_0^1 f_1 f_2\,dx$$

Hence $\delta_2(x_1) = \delta_1(x_2)$

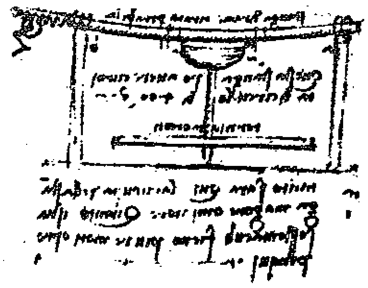
(8.66)





From Table 8.1

$$v = \frac{w_0 x}{24EI}\left(L^3 - 2Lx^2 + x^3\right)$$



Potter's wheel for making mirrors with large focal length - Cod. Arund., fol. 84 v

ρ = radius of curvature

$$\frac{1}{\rho} \approx \frac{d^2 v}{dx^2} = \frac{w_0 x (L-x)}{2EI}$$

$$w_0 \,(N/m) = \gamma A \quad : \gamma \,(N/m^3)$$

$$= 43 N/m$$

$$\frac{1}{\rho} = \frac{w_0 x}{2EI}(L-x) \quad ; \text{at } x = \frac{L}{2} \quad \frac{1}{\rho} = \frac{w_0 L^2}{8EI}$$

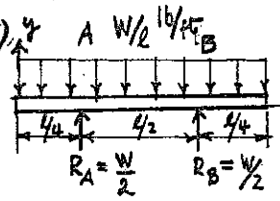
WITH $L = 2.5$ m

$E = 117 \times 10^9$ N/m^2

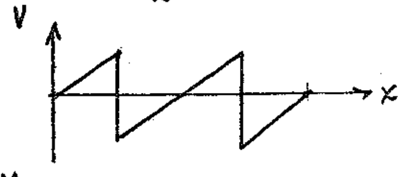
$I = 1.92 \times 10^{-8}$ m^4

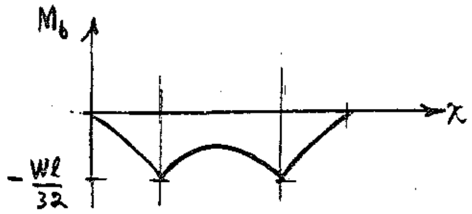
$$\rho = 67 m$$

(8.67)



Assume that the load on the rollers is the evenly distributed weight of the log.

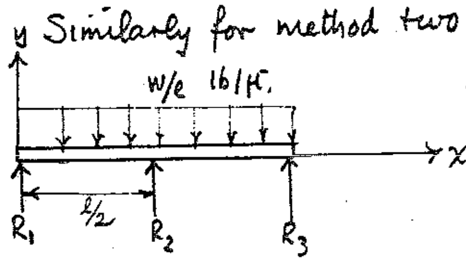




From the bending moment diagram, the $(M_b)_{max}$ occurs at $x = \frac{l}{4}, \frac{3l}{4}$ the location of the roller,

$\therefore$, for the first method, the log would break at one of the rollers.

Similarly for method two:



$\Sigma F_y = 0 \quad R_1 + R_2 + R_3 = 0$

From symmetry $R_1 = R_3$

$\therefore \quad 2R_1 + R_2 = W \quad \cdots\cdots$ (I)

$\Sigma M = 0, \quad R_1 x - \frac{W}{l}\frac{x^2}{2} + R_2\langle x - \frac{l}{2}\rangle^1 = EI\frac{d^2v}{dx^2}$

B.C. (1) $x = 0 \quad v = 0$

(2) $x = l/2 \quad v = 0$

(3) $x = l \quad v = 0$

$$EI\frac{dv}{dx} = \frac{R_1x^2}{2} - \frac{Wx^3}{6l} + \frac{R_2}{2}\langle x - l/2\rangle^2 + C_1$$

$$EIv = \frac{R_1x^3}{6} - \frac{Wx^4}{24l} + \frac{R_2}{6}\langle x - l/2\rangle^3 + C_1x + C_2$$

Applying B.C. (1) $\longrightarrow \quad C_2 = 0$

" " " (2) $\longrightarrow \quad C_1 = -R_1\frac{l^2}{24} + \frac{Wl^2}{16\times 12}$

$$EIv = \frac{R_1x^3}{6} - \frac{Wx^4}{24l} + \frac{R_2\langle x - l/2\rangle^3}{6} - \frac{R_1l^2x}{24} + \frac{Wl^2x}{16\times 12}$$

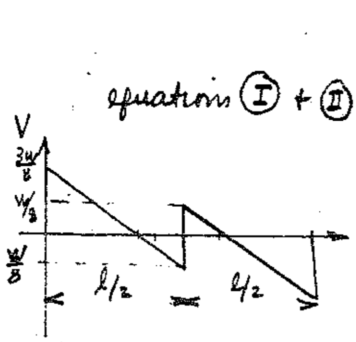
Applying B.C. (3)

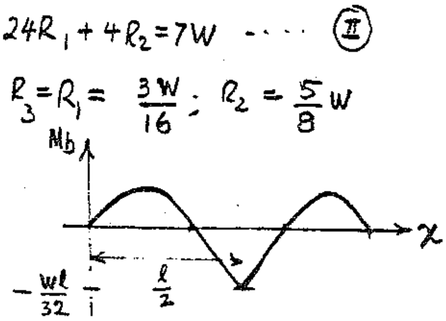
$$\frac{R_1l^3}{6} - \frac{Wl^3}{24} + \frac{R_2}{6}\cdot\frac{l^3}{8} - \frac{R_1l^3}{24} + \frac{Wl^3}{16\times 12} = 0$$

$\therefore 24R_1 + 4R_2 = 7W \quad \cdots$ (II)

equations (I) + (II) gives $R_3 = R_1 = \frac{3W}{16}; \; R_2 = \frac{5}{8}W$

From Prob. (8.16) with suitable change in dimensions $(M_b)_{max} = \frac{Wl}{32}$ same as part (a)





the $(M_b)_{max}$ is at the center roller. $\therefore$ the column breaks "upon the middle support".